J. Plasma Phys. (2026), *vol.* 92, E66 © The Author(s), 2026. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives licence (https://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided that no alterations are made and the original article is properly cited. The written permission of Cambridge University Press or the rights holder(s) must be obtained prior to any commercial use and/or adaptation of the article.
doi:10.1017/S002237782610155X

Power and particle exhaust for the ARC fusion power plant

Thomas H. Eich[1,#], Thomas A.J. Body[1,#], Tom P. Looby[1], Sean B. Ballinger[1], Alexander J. Creely[1], Jon C. Hillesheim[1], Philip B. Snyder[1], Nathan T. Howard[2], Rebecca Masline[2], Michael R.K. Wigram[2] and George R. Tynan[2]

[1]Commonwealth Fusion Systems, Devens, MA, USA
[2]MIT Plasma Science and Fusion Center, Cambridge, MA, USA

Corresponding authors: Thomas H. Eich, teich@cfs.energy; Thomas A.J. Body, tbody@cfs.energy

(Received 17 October 2025; revision received 30 January 2026; accepted 3 March 2026)

To successfully show that fusion is an attractive energy source, the ARC^{TM} fusion power plant will need to operate with a robust, integrated power and particle exhaust solution. To maximise ARC's fusion power output while avoiding excessive erosion of the plasma-facing components, we will need to radiatively dissipate most of the power crossing the last-closed flux surface, injecting radiating impurities such as argon or neon to access divertor detachment. Divertor detachment will need to be integrated with a high-performance core plasma, and with efficient impurity pumping to prevent the accumulation of helium ash in the core. To access and control detachment in high-performance plasmas, we have designed ARC with up–down-symmetric divertors, with secondary X-points in long, tightly baffled outer legs. Using a core-edge modelling workflow, we predict that with this divertor design, ARC will access detachment with modest argon seeding in the divertor ($c_{Ar,div} \sim 0.9\,\%$), which should have minimal impact on the core ($\Delta Z_{eff,core} < 0.5$) for reasonable argon enrichment ($c_{Ar,div}/c_{Ar,core} = 6$). Due to the high predicted divertor neutral pressure ($p_{div} \sim 20$ Pa), we predict that ARC will sufficiently pump helium to limit ash accumulation in the core ($c_{He,core} < 2\,\%$) for a helium enrichment of $c_{He,div}/c_{He,core} = 0.4$. ARC's divertor design is expected to increase the stability of a detachment front in the outer divertor leg, helping to prevent divertor reattachment during smaller heat-flux transients such as scrape-off-layer filaments associated with the quasi-continuous exhaust regime. However, this buffering will not be sufficient to prevent divertor reattachment during large type-I edge-localised modes (ELMs), and as such these will need to be avoided on ARC. Experiments on SPARC will be used to select an integrated scenario which avoids or mitigates type-I-ELMs while maintaining access to detachment, good core fusion performance and sufficient impurity exhaust. SPARC experiments will also be used to finalise ARC's divertor design, by studying the impact of magnetic and first-wall geometry on detachment stability, impurity enrichment and neutral baffling under conditions similar to those expected for ARC. In conclusion, our analysis finds that ARC will have a viable power and particle exhaust solution which is

[#]T.H. Eich and T.A.J. Body are co-first authors of this work.
*Nomenclature is defined in Appendix C.
†SPARC® and ARC^{TM} are trademarks of Commonwealth Fusion Systems®.

compatible with high-power operations, and this solution will be validated in experiments on SPARC.

Key words: fusion plasma

1. Introduction

The ARC fusion power plant is designed to reliably and cost-effectively produce 400 megawatts of net electricity from fusion (Hillesheim *et al.* 2026). To achieve this, the core plasma of ARC will be heated to over 100 million degrees Celsius – more than 10 times hotter than the centre of the Sun. This high-temperature plasma is separated by a little over a metre ($a_{geo} \sim 1.18$ m) from the nearest solid plasma-facing components (PFCs) of the tokamak, which must be kept below a few thousand degrees Celsius (Ueda *et al.* 2014; Linke *et al.* 2019). To maintain the temperature gradient from core to edge, a strong magnetic field ($B_{axis} \sim 11.4$ T) will be used to confine the plasma and 250 MW of heating power (mostly alpha heating from fusion reactions) will be continuously added into the core (Hillesheim *et al.* 2026). Due to stiff ion-temperature-gradient turbulence, the resulting temperature gradient is nevertheless predicted to be relatively flat throughout most of the confined region – and as a consequence, ARC's fusion power output will strongly depend on gradients in the edge pedestal region (Howard *et al.* 2026). To develop steep gradients in the edge, we need to provide sufficient power to develop and sustain an edge transport barrier in a high-confinement H-mode (Wagner *et al.* 1982; Wagner 2007). Based on empirical scalings (Martin, Takizuka & the ITPA CDBM H-Mode Threshold Database Working Group 2008; Vincenzi *et al.* 2025; Delabie *et al.* 2026), the threshold power is estimated to be around ∼100 MW for the ARC flattop (Hillesheim *et al.* 2026). Although these threshold-power scalings were derived in terms of the input power minus the change in stored energy, results from highly radiating plasmas indicate that this threshold power needs to reach the separatrix (Hughes *et al.* 2011; Solano *et al.* 2021) – with performance continuing to improve as the power crossing the separatrix is increased above the threshold by a factor of up to $P_{sep}/P_{L-H} \sim 1.5$ (Hughes *et al.* 2011).

Once the power crosses the separatrix it reaches the scrape-off-layer (SOL) where magnetic field lines intersect the solid walls of the divertor (Pitcher & Stangeby 1997). Since the power is being carried by plasma particles, it streams much more quickly along than across the magnetic field lines, and this is true also in the SOL. As a consequence, without additional power dissipation, most of the power crossing the separatrix arrives at the divertor surface in a narrow channel (Eich *et al.* 2011; Goldston 2012; Eich *et al.* 2013), leading to a concentrated heat flux that can cause melting or cracking of the divertor surface (Linke *et al.* 2019). Even if the power dissipation is sufficient to avoid melting or cracking, high-energy plasma ions can sputter atoms from the divertor surface, over time leading to material erosion (Hakola *et al.* 2021) and an increased risk of UFO disruptions (Maget *et al.* 2025). In addition, typical high-performance H-modes periodically release bursts of energy called edge-localised modes (ELMs) (Zohm 1996; Wilson *et al.* 2006). Large type-I ELMs can melt and displace significant amounts of material from the divertor surface of existing devices such as JET (Coenen *et al.* 2015), and they are predicted to have even higher-energy fluences in tokamaks with higher pedestal pressures (Eich *et al.* 2017), making them a significant concern for ARC. As well as having to deal

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

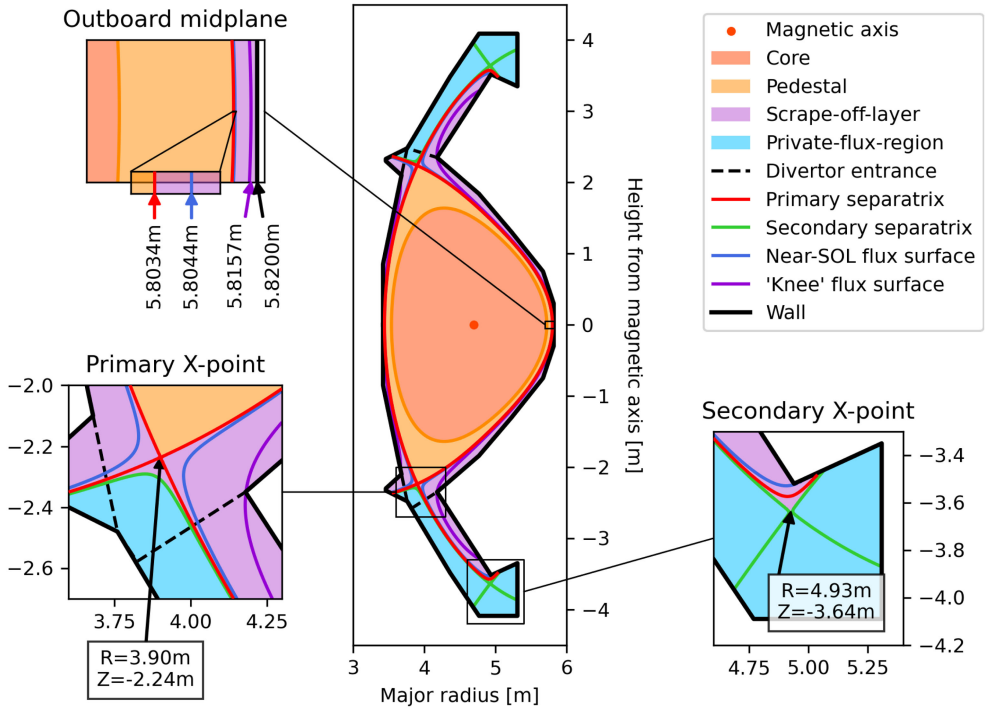


FIGURE 1. A poloidal cross-section of the ARC V3A design, highlighting key regions and terminology used in this paper. Note that the engineering design is not final.

with power exhaust, the edge and divertor must also exhaust helium ash and other impurity particles at a sufficient rate to prevent fuel dilution in the core (Reiter *et al.* 1991).

Protecting PFCs from continuous and transient heat fluxes while maximising core fusion performance has been recognised as a critical challenge for several fusion power plant designs (Zohm *et al.* 2021); Asakura *et al.* 2021, 2023). As a result, these topics have been a key research focus for the international fusion community (Federici *et al.* 2001), and significant progress has been made in the development of highly dissipative and small- or no-ELM scenarios (Campbell *et al.* 2025; Fenstermacher *et al.* 2025; Krieger *et al.* 2025). Experiments on existing tokamaks have shown that, through the controlled injection of impurities such as neon or argon, power can be dissipated from the primary heat exhaust channel and spread across a much larger area of the wall. Impurity seeding has proven to be highly effective at protecting against steady-state heat fluxes on several tokamaks (Kallenbach *et al.* 2011; Xu *et al.* 2020) and may also reduce transient heat fluxes (Komm *et al.* 2023). However, due to their higher mass and charge, impurities can significantly increase the rate at which particles are sputtered from the PFCs (van Rooij *et al.* 2013). To prevent sputtering and erosion of the divertor targets, additional power dissipation can be used to cool the SOL plasma to the point where it recombines into neutral gas and detaches from the divertor targets (Krasheninnikov & Kukushkin 2017). Detachment greatly reduces divertor sputtering, but needs to be controlled to prevent excessive edge cooling which can reduce confinement or trigger disruptions (Lipschultz 1987; Li *et al.* 2021). Active detachment control has been

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

successfully demonstrated on several tokamaks including TCV, EAST, KSTAR and DIII-D (Eldon *et al.* 2021, 2022; Ravensbergen *et al.* 2021; Wang *et al.* 2022), and significant effort has gone into improving passive detachment stability and compatibility with core performance through optimisation of the divertor magnetic geometry (Soukhanovskii 2017; Verhaegh *et al.* 2025) and physical neutral baffling (Wang *et al.* 2021; Gao *et al.* 2023; Maurizio *et al.* 2024; Meng *et al.* 2024).

Detachment protects against sputtering, melting and cracking of the divertor targets due to the near-steady-state power crossing the separatrix, but large power transients such as type-I-ELMs can burn through the neutral gas layer (Flanagan *et al.* 2025). To reduce the size of heat-flux transients, scenarios have been developed which produce pedestal pressures approaching those found in typical type-I-ELMy H-modes while avoiding or suppressing large ELMs – including the quasi-continuous exhaust (QCE/type-II ELM) regime (Stober *et al.* 2001; Faitsch *et al.* 2023, 2025; Saibene *et al.* 2005), grassy ELM regime (Kamada *et al.* 2000), enhanced D-alpha regime (Greenwald *et al.* 1999; Hubbard *et al.* 2001), ELM-suppressed X-point radiator regime (Bernert *et al.* 2020), quiescent H-mode (Burrell *et al.* 2002) and wide-pedestal quiescent H-mode (Burrell *et al.* 2020; Ernst *et al.* 2024), M-mode (Garcia *et al.* 2024), I-mode (Whyte *et al.* 2010) and super-I-mode (Song *et al.* 2023), as well as scenarios where type-I-ELMs are suppressed via resonant magnetic perturbations (RMPs) (Moyer *et al.* 2005) or pellet pacing (Lang *et al.* 2004). In parallel, helium exhaust experiments have demonstrated that helium exhaust out of the confined region is typically sufficient for a fusion power plant – while removing and pumping helium from the SOL remains challenging (Wade *et al.* 1995; Bosch *et al.* 1997; Sakasai *et al.* 1999) and the focus of research (Zito *et al.* 2023; Masline, Wigram & Whyte 2025). These results demonstrate that solutions can be found for heat and particle exhaust, albeit not always simultaneously.

The challenge for ARC is to develop an integrated solution that handles the continuous and transient heat fluxes (projected to be higher than those in most existing devices), provides sufficient helium exhaust, maintains high pedestal pressures and remains compatible with long pulses, a high duty cycle and significant material degradation due to neutron fluxes. To find integrated solutions for the ARC design as it evolved through design iterations, we focused on medium-fidelity models either informed by or validated against experimental results from the fusion community. In this work, we present the results of this medium-fidelity workflow for the ARC V3A design point, while results from ongoing higher-fidelity SOLPS-ITER and SOLEDGE3X transport modelling will be presented in later publications. We first predict the continuous and transient heat fluxes for ARC – in §§ 2.1 and 2.3 respectively – applying scalings developed from multi-machine databases. Based on these results, we find that ARC will need to operate with detached divertors to prevent excessive erosion of the divertor targets, and that large type-I-ELMs will need to be either avoided or mitigated. In § 2.4 we test whether we can use detachment to buffer against ELMs, and find that – although type-I-ELMs are likely to burn through the detachment front – smaller ELMs or filaments should be able to be buffered. Based on this, we designed ARC with X-point target divertors to allow for easier detachment access and control. We discuss recent research on passive detachment stability in § 3.1, and how the X-point target configuration aims to maximise detachment stability in 3.2. We then show the divertor design developed for ARC in § 3.3, and discuss ongoing work to optimise this design in § 3.4.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

To show that this divertor design leads to a viable heat exhaust solution for ARC, in § 4 we introduce a workflow to assess whether a given core scenario is compatible with detachment. For a range of coupled core and pedestal solutions identified via integrated modelling, we use the 'extended Lengyel' model to predict the concentration of argon which we would need to inject into the divertor in order to access detachment, and then calculate the impurity enrichment required to reach a viable core-edge integrated scenario. These results are generally favourable – finding that even at ARC's maximum fusion performance, we should be able to access detachment with an argon concentration of ∼0.9 % in the divertor. We then check whether this operational point also has a viable particle exhaust rate in § 5, and show that due to the high predicted divertor neutral pressure of ∼20 Pa ARC should be able to limit the core helium concentration to a few per cent. Finally, we discuss several high-performance, type-I-ELM-free regimes being considered for ARC in § 6. Focusing on the QCE regime in § 6.1, we find that at detachment onset, ARC should have sufficient shaping and edge-pressure gradients to access this regime according to a model proposed by Dunne *et al.* (2024) – but this result is called into question since the α_t turbulence parameter does not meet an access criterion empirically determined by Faitsch *et al.* (2023). In summary, our analysis finds that ARC should be able to achieve good core fusion performance while maintaining detachment, limiting core helium accumulation for reasonable impurity enrichment values and potentially intrinsically avoiding type-I-ELMs.

2. The challenge of power exhaust for ARC

Fusion power plants need to find a balance between maximising their fusion performance and protecting their PFCs from power exhaust. Tokamaks operating in a high-performance H-mode need to tackle two interconnected challenges: dissipating the inter-ELM heat exhaust and avoiding or mitigating large type-I ELMs. These challenges are already faced by existing tokamaks, which need to carefully design experiments to avoid damaging the device walls and minimise contamination of the core plasma. Compared with these devices, fusion power plants will operate at higher power and for orders of magnitude more cumulative 'plasma-on' time, making the need to identify a power exhaust solution even more critical. Finding a viable power exhaust solution and integrating this with a high-performance core will be crucial to the success of ARC. A useful metric for the trade-off between achieving benign SOL conditions and core performance is the separatrix turbulence parameter

$$\alpha_t \simeq 3 \times 10^{-18} R_{axis} q_{cyl}^2 \frac{n_{e,u}}{{T_{e,u}}^2} Z_{eff,u} \approx \frac{q_{cyl} \cdot \nu^*_{e,edge}}{100}, \tag{2.1}$$

which gives the strength of ballooning instabilities near the separatrix, relative to drift waves (Eich *et al.* 2020). The α_t parameter can be thought of as a normalisation of the separatrix collisionality $\nu^*_{e,edge}$ (Stangeby 2018), with values typically ranging from ∼0.1 to ∼1.0 for most operating tokamaks. Higher values of α_t tend to indicate more benign power exhaust conditions (when operating in H-mode) for several reasons; higher values of normalised collisionality typically reduce the fraction of pedestal stored energy lost in an ELM (Loarte *et al.* 2003), small-ELM or filamentary H-modes have been found to require a minimum value of the normalised collisionality (Wolfrum *et al.* 2011; Labit *et al.* 2019; Faitsch *et al.* 2023; Miller *et al.* 2025), and (as we discuss in Appendix B) the impurity concentration required to access detachment scales with $c_z \propto {\alpha_t}^{-2}$ (Goldston,

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

Reinke & Schwartz 2017; Reinke 2017). However, higher values of α_t are also correlated with reduced confinement (Eich *et al.* 2020), higher levels of cross-field transport (LaBombard *et al.* 2005)[1] and eventually either lead to an L-mode back transition or to the H-mode density limit (Goldston 2015; Manz, Eich & Grover 2023; Manz 2025). Therefore, the highest fusion performance in ARC will be found at the minimum value of α_t that leads to a robust power exhaust solution with stable detachment and without type-I-ELMs.

In the remainder of this section, we predict the continuous and transient heat fluxes for ARC. In § 2.1, we show that continuous divertor detachment throughout the flat-top period will be essential to mitigate risks such as tile cracking from thermal stresses, melting from bulk heating and erosion and core pollution from sputtering. In § 2.3, we show that continuous operation with unmitigated type-I ELMs is infeasible, due to increases in the divertor ELM energy density caused by the significantly higher pedestal top pressure (when compared with operating tokamaks) (Eich *et al.* 2017).

2.1. *Continuous heat fluxes and the need for detachment*

To demonstrate why detachment is required on ARC, we first need to estimate how much heat is entering the SOL during the quasi-steady-state inter-ELM period, and determine the impact this will have on the divertor. The heat flux in the SOL decays exponentially as we move away from the separatrix, with a characteristic scale length $\lambda_{q,u}$. The parallel heat-flux density in the first $\lambda_{q,u}$ of the SOL, when measured at the outboard midplane, is expected to be

$$q_{\parallel,u} = \frac{P_{sep}(1-1/e)\,f_{odiv}}{2\pi(R_{axis}+a_{geo})\lambda_{q,u}}\frac{B_{tot,u}}{B_{pol,u}}, \tag{2.2}$$

where P_{sep} is the total power crossing the separatrix, of which we assume a fraction f_{odiv} is directed towards one of the outer divertors. Approximately $f_{odiv} \sim 2/3$ of this usually flows towards the outer divertor in tokamaks with a single divertor, but for SPARC and ARC we will assume $f_{odiv} \sim 1/2$ since both devices have two outer divertors (Brunner *et al.* 2018*a*). Of the power directed to one of the divertors, we assume that $1-1/e = 0.63$ of the power is transported in the first $\lambda_{q,u}$. This power is assumed to be spread over a ring with a circumference $2\pi(R_{axis}+a_{geo})$, where R_{axis} is the major radius and a_{geo} is the minor radius. For our initial analysis we use regression 9 from Eich *et al.* (2013) to estimate the width of the ring

$$\lambda_{q,u}[\mathrm{mm}] = 0.70\,B_{axis}[\mathrm{T}]^{-0.77} q_{95}^{1.05} P_{sep}[\mathrm{MW}]^{0.09}, \tag{2.3}$$

while in § 4 we calculate $\lambda_{q,u}$ using (4.2) to account for turbulent broadening at higher α_t values. Finally, we multiply by the magnetic field-line pitch at the outboard midplane $(B_{tot,u}/B_{pol,u})$ to project the heat flux from the poloidal direction to the parallel direction.

In table 1 we compare the heat flux entering the SOL for several existing tokamaks with SPARC and ARC, taking typical values for P_{sep} and other machine parameters. Due to their high magnetic fields and plasma currents, both SPARC and ARC are predicted to require higher P_{sep} to stay in H-mode than on existing tokamaks (Reinke 2017; Hillesheim *et al.* 2026), and this power must be exhausted in a

[1] LaBombard et al. (2005) uses α_d, which can be related to $\alpha_t \propto 1/\alpha_d^2$ using from (14) of Eich et al. (2020)

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

	Alcator C-Mod	ASDEX Upgrade	JET	SPARC	ARC V3A
R_{axis} (m)	0.67	1.6	3	1.9	4.6
a_{geo} (m)	0.22	0.5	0.95	0.57	1.2
B_{axis} (T)	5.4	2.5	3	12	11
I_p (MA)	0.8	1	2.5	8.7	12
$\langle m_i \rangle$ (amu)	2	2	2	2.5	2.5
κ_{areal}	1.5	1.6	1.6	1.8	1.6
δ_{95}	0.45	0.35	0.3	0.3	0.36
f_{odiv}	0.67	0.67	0.67	0.5	0.5
q_{95}	5.7	4.5	4.3	3.3	3.2
P_{sep} (MW)	2	10	20	28	120
P_{sep}/P_{L-H}	1.1	3.6	2	1	1.2
$\langle n_e \rangle$ (10^{20}m^{-3})	2	0.8	0.8	2.9	2.4
$\langle n_e \rangle / n_{GW}$	0.38	0.63	0.91	0.34	0.89
$\lambda_{q,u}$ (mm)	1.3	2.1	1.8	0.48	0.56
$q_{\parallel,u}$ (GW/m^2)	0.67	0.74	0.83	3.9	8.5

TABLE 1. Representative values used to calculate the unmitigated heat-flux density $q_{\parallel,u}$ for Alcator C-Mod (Brunner *et al.* 2018*b*; Greenwald *et al.* (2014), ASDEX Upgrade (Zohm *et al.* 2024), JET (Kappatou *et al.* 2025; Rimini, JET Contributors & the EUROfusion Tokamak Exploitation Team 2025), SPARC (Creely *et al.* 2020; Kuang *et al.* 2020; Body, Hasse & Creely 2023) and ARC (Hillesheim *et al.* 2026).

narrower $\lambda_{q,u}$. As a result, both SPARC and ARC will have higher heat fluxes flowing into the near SOL than what has been seen on existing devices.

Once in the SOL, plasma will stream along the magnetic field line towards the divertor targets. Some fraction of the heat flux f_{pow} will be dissipated from the near-SOL via radiation or due to cross-field transport, or distributed over a larger area via flux expansion, resulting in a reduced heat flux reaching the divertor target

$$q_{\parallel,t} = q_{\parallel,u}(1 - f_{pow})\frac{R_u}{R_t}, \tag{2.4}$$

which corresponds to a wall-normal heat flux

$$q_{\perp,t} = q_{\parallel,t} \sin(\theta_\perp), \tag{2.5}$$

where $\theta_\perp$ is the angle of incidence between the fieldline and the wall. To allow for machining and assembly tolerances, we assume $\theta_\perp \sim 2.5°$ for the following analysis.

If the wall-normal heat flux $q_{\perp,t}$ exceeds the rate at which heat flows out of the divertor, the divertor temperature will increase. Sustained heating can lead to top-surface melting of the divertor targets, or cracking due to thermal gradients. The dynamics of target melting and cracking is complex, and depends on the exact design and history of the divertor targets. We take $\sim$10 MW m^{-2} as an order-of-magnitude limit at which melting and cracking starts to become a risk. To stay below this limit, we need to dissipate at least 97 % of the 8.5 GW m^{-2} entering the near-SOL, which corresponds to a SOL radiation fraction of $f_{rad} = 1 - \lambda_{INT}/\lambda_{q,u}(1 - f_{pow}) = 88\,\%$ if we assume that $\lambda_{INT}/\lambda_{q,u} = 4$ due to cross-field transport in the divertor (see § 4.3). We estimate the sheath-entrance electron temperature using the two-point-model

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

(Stangeby 2018)

$$T_{e,t} = \frac{8\langle m_i \rangle}{\gamma_{sh}^2} \frac{q_{\parallel,t}^2}{(1 - f_{mom}) n_{e,u} (T_{e,u} + T_{i,u})}, \tag{2.6}$$

where $\langle m_i \rangle$ is the average ion mass, $\gamma_{sh} \approx 7.5$ is the sheath heat transmission factor, $1 - f_{mom}$ is the momentum loss, $n_{e,u}$ is the upstream electron density and $T_{e,u}$ and $T_{i,u}$ are the upstream electron and ion temperatures. We assume $T_{e,u} = T_{i,u}$ and calculate

$$T_{e,u} = \left(T_{e,t}^{7/2} + \frac{7}{2} \frac{q_{\parallel,u} L_\parallel}{\kappa_{e,0}} \right)^{2/7} \tag{2.7}$$

from Spitzer–Harm, using $L_\parallel = 120\,\mathrm{m}$ (which is slightly less than the $L_\parallel = 170\,\mathrm{m}$ from the reference equilibrium – see discussion in § 3). If we limit the target heat flux to $\sim 10\,\mathrm{MW\,m^{-2}}$, the sheath-entrance electron temperature will be $\sim 2\,\mathrm{eV}$. This implies that the divertor target will need to be detached to avoid target melting and cracking limits.

A further argument for why detachment is necessary for ARC comes from calculating the net erosion of the divertor targets. To radiate enough power to stay below melting and cracking limits, we will need to inject noble-gas impurities such as neon or argon into the divertor. These impurities have more electrons than the deuterium and tritium fuel ions, which makes them efficient at radiating power at the temperatures predicted for ARC's SOL. However, due to their higher masses, these impurities will also increase the sputtering rate from the divertor compared with a pure deuterium-tritium plasma, which introduces several risks for ARC. Sputtering increases the concentration of tungsten in the core (especially if the sputtered tungsten originates from the main chamber (Dux *et al.* 2009; Angioni 2025)). Once in the core, tungsten very efficiently radiates power from the confined region, which can reduce the energy confinement in the core or even trigger a radiative collapse (Ostuni *et al.* 2022). Sputtered tungsten can also be redeposited in large, poorly attached flakes, which can trigger UFO disruptions when they detach from the walls (Maget *et al.* 2025). Over the lifetime of ARC's divertor, sustained sputtering from a localised source such as the divertor strike points will also lead to cumulative material erosion. As a rough estimate for how significant erosion might be if we operated ARC with a dissipative-but-not-detached divertor, we calculate what the direct sputtering would be if we increased the sheath-entrance electron temperature to $\sim 10\,\mathrm{eV}$ (corresponding to a target heat flux of $\sim 20\,\mathrm{MW\,m^{-2}}$). Using the sputtering yields shown in figure 2, we see that at $T_{e,t} \sim 10\,\mathrm{eV}$ deuterium and tritium ions are not expected to contribute significantly to sputtering, but that argon ions will sputter 1 tungsten atom for every ~ 8 argon ions reaching the target. From the two-point model (Stangeby 2018), we estimate a main ion particle flux of $6 \times 10^{25}\,\mathrm{m^{-2}s^{-1}}$ to the target. If we assume a $\sim 0.5\,\%$ argon concentration is needed for power dissipation, we find a direct sputtering rate of $\sim 4 \times 10^{22}\,\mathrm{m^{-2}s^{-1}}$. Taking typical values for tungsten's mass (183.8 amu) and density ($19.6\,\mathrm{g/cm^3}$), this would correspond to a net erosion of $\sim 600\,\mathrm{nm/s}$ or $\sim 20\,\mathrm{m}$ per full power year. Of this, we expect about $\sim 94\,\%$ of the sputtered tungsten atoms to be redeposited close to their point of origin (Brezinsek *et al.* 2019). Accounting for redeposition, the gross erosion rate is predicted to be $\sim 1\,\mathrm{m}$ per full power year, which is much more than the feasible thickness of the tungsten armour in the divertor. To limit erosion to reasonable levels, we will operate ARC with detached divertors.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

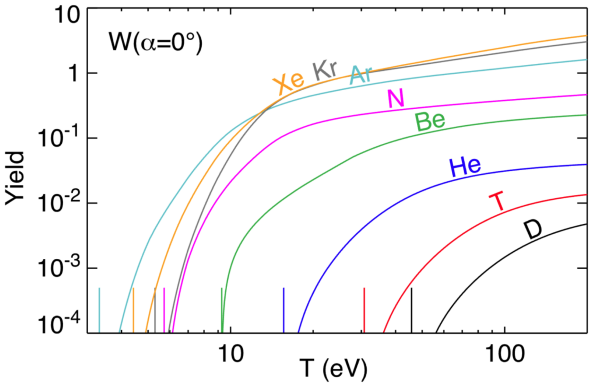


FIGURE 2. The number of tungsten atoms sputtered per incident ion (sputtering yield) as a function of sheath-entrance electron temperature, for an ion flux perpendicular to the tungsten surface. The sputtering yield is calculated in terms of sheath-entrance temperatures by assuming the impact energy is approximately $E_{impact} = 2T_e + 3ZT_e$. Reproduced with permission from Neu *et al.* 2026.

By reducing the target electron temperature to $\lesssim$ 2eV the sputtering yield will drop by more than three orders of magnitude (see figure 2), and the incident particle flux will drop once the plasma detaches (Krasheninnikov & Kukushkin 2017). Detachment will therefore reduce the erosion rate in the divertor to negligible levels, which is consistent with results from higher-fidelity modelling (Lasa *et al.* 2026).

2.2. *Other sources of erosion*

Once the plasma is detached and divertor sputtering is suppressed, other sources of sputtering will become the dominant contributors to erosion. Charge-exchange neutrals (Verbeek *et al.* 1998; Yoshida & Hirooka 1998; Baumann *et al.* 2026), α particles (Bauer *et al.* 1979; Tobita *et al.* 2003), radio-frequency sheaths (Bobkov *et al.* 2011; Chomiczewska *et al.* 2024) and filamentary transport into the far-SOL (Redl *et al.* 2023; Perillo *et al.* 2025) will all contribute to sputtering and erosion, primarily from the main chamber (outside of the divertor). These sputtering sources may lead to significant amounts of material migrating from the main chamber to the detached divertor (Stangeby *et al.* 2022), both contributing to erosion and producing significant amounts of slag which can increase the risk of disruptions (Stangeby *et al.* 2022; Maget *et al.* 2025). To assess the erosion rate due to charge-exchange neutrals, wide-grid transport modelling with SOLEDGE3X (Rivals *et al.* 2022) is being performed for ARC. This modelling will help to inform the design of the vacuum vessel and to determine the replacement schedule or whether in-situ maintenance will be needed.

2.3. *Transient heat fluxes and the need for small-ELM scenarios*

In addition to protecting the divertors from the continuous heat flux predicted for ARC, we also need to protect against heat-flux transients which cause the power crossing the separatrix to rapidly increase for a short amount of time. Of these, we focus on ELMs to show why ARC will need to operate in a scenario which avoids large type-I-ELMs. Type-I-ELMs are a natural consequence of operating in a typical

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

high-performance H-mode. To understand why ELMs occur and to discuss how they might be avoided on ARC, it is helpful to first introduce a simplified model of the physics which drives them (Snyder *et al.* 2011, 2012). In our simple picture, once an edge transport barrier is formed much of the turbulent radial heat and particle fluxes in the edge are suppressed, and pedestal gradients typically rise until they are constrained by the onset of the kinetic ballooning mode. With reduced radial fluxes through the pedestal, the pedestal width and height increase continuously, in turn driving an increasing amount of bootstrap current. This process continues until a non-local magnetohydrodynamics limit is reached: either a pressure-driven medium-to-high toroidal-mode-number ballooning mode, a current-driven low toroidal-mode-number kink-peeling mode, or a coupled peeling-ballooning mode (Snyder *et al.* 2002; Viezzer 2018). Once this limit is reached, the pedestal energy is rapidly ejected towards the divertor targets as an ELM (or, in certain circumstances, the non-local mode can saturate leading to quiescent H-mode operation). Despite being short-lived, these ELM events can deposit significant amounts of energy into the divertor, leading to top-surface melting, cracking and a significant (even dominant) fraction of the total sputtering (Brezinsek *et al.* 2019; Krieger *et al.* 2025).

To quantitatively show why type-I-ELMs need to be avoided on ARC, we use a model for the peak energy fluence to the divertor from Eich *et al.* (2017). The model assumes that, during an ELM, magnetic field lines connect the pedestal top to the divertor targets, resulting in a rapid energy loss due to parallel transport. The total energy available is estimated from the pedestal top pressure $p_{e,ped}$ and plasma volume near the pedestal top, which is assumed to be evenly distributed over the effective area of the divertor targets. For simplicity we assume that $T_{e,ped} = T_{i,ped}$, $Z_{eff,ped} = 1$, $R_t = R_{ped} = R_{axis}$ to arrive at a simple equation for the energy fluence

$$\epsilon_{\parallel} \cong C \cdot 6\pi \cdot p_{e,ped} \cdot R_{axis} \cdot q_{edge}, \tag{2.8}$$

$$q_{edge} = \sqrt{\frac{1+\kappa_{95}^2}{2}} \frac{a_{geo}}{R_{axis}} \frac{B_{tor,u}}{B_{pol,u}} \tag{2.9}$$

$$= \sqrt{\frac{1+\kappa_{95}^2}{2}} \frac{a_{geo}}{R_{axis}+a_{geo}} \frac{B_{axis}}{B_{pol,u}}. \tag{2.10}$$

Despite its simplicity, (2.8) with $C = 1$ reproduces the minimum $\epsilon_{\parallel}$ observed on MAST, ASDEX Upgrade and JET (Eich *et al.* 2017), while the largest $\epsilon_{\parallel}$ was within a factor of 3 of (2.8) for experimental data spanning more than two orders of magnitude of $\epsilon_{\parallel}$. Therefore, we expect the type-I-ELM energy fluence to be described by (2.8) for C between 1 and 3. At the high fusion performance point discussed in § 4, the predicted pedestal pressure for ARC is $p_{tot,ped} = 280\,\mathrm{kPa}$ (evaluated at one half-width in from the centre of the pedestal tanh fit – this is approximately 28 % lower than the pedestal top value, evaluated two half-widths in from the centre of the pedestal tanh fit, used as a boundary condition for core turbulence simulations in Howard *et al.* (2026)). This gives an electron pressure of $p_{e,ped} = 140\,\mathrm{kPa}$ since we have assumed $T_{e,ped} \approx T_{i,ped}$. Then, taking $\kappa_{95} = 1.5$ and $B_{pol,u} = 2\,\mathrm{T}$ from the ARC reference equilibrium and values from table 1, we calculate $q_{edge} = 1.5$ using (2.10). Therefore, we expect a minimum type-I-ELM energy fluence ($C = 1$) of

$$\epsilon_{\parallel} \cong 6\pi \cdot 140\,\mathrm{kPa} \cdot 4.62\,\mathrm{m} \cdot 1.5 = 18\,\mathrm{MJ\,m^{-2}}, \tag{2.11}$$

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

which corresponds to target-normal energy fluence of

$$\epsilon_\perp \cong \epsilon_\parallel \sin(2.5^\circ) = 0.8\ \mathrm{MJ\ m^{-2}} \tag{2.12}$$

and a maximum energy fluence ($C = 3$) of $\epsilon_\perp = 2.4\ \mathrm{MJ\ m^{-2}}$. To calculate the impact of a type-I-ELM on the divertor targets, we use the analytical 'free streaming model' (FSM) (Fundamenski, PITTS & JET EFDA Contributors 2006). This model describes the energy and particle transport of ELM filaments from the pedestal to the divertor targets, and it has been validated on several tokamaks (Eich *et al.* 2009; Guillemaut *et a.* 2018). Using this model for ARC, we assume the ELM transports ions to the target over a characteristic time scale

$$\tau_{FSM} = L_\parallel / \sqrt{(T_{e,ped} + T_{i,ped})/\langle m_i \rangle} \tag{2.13}$$

set by the parallel connection length and the plasma sound speed at the pedestal top. Using $L_\parallel = 120$ m (see § 3) and taking $T_{e,ped} = T_{i,ped} = 4.2$ keV, we find $\tau_{FSM} = 210\ \mu$s. The values of τ_{FSM} and $\epsilon_\perp$ can be used to define a functional form for the ELM energy deposition via an adapted form of the FSM as defined in Eich *et al.* (2009)

$$q_{ELM}(t) = 2 \cdot \frac{2}{3}\frac{\epsilon_\perp}{\sqrt{\pi}} \cdot \left[1 + \left(\frac{\tau_{FSM}}{t}\right)^2\right] \cdot \frac{\tau_{FSM}}{t^2} \cdot \exp\left[-\left(\frac{\tau_{FSM}}{t} - H_{fl} M_{\parallel,tor}\right)^2\right], \tag{2.14}$$

where H_{fl} is a sign variable (-1 or 1) corresponding to the plasma helicity and $M_{\parallel,tor}$ is the mach number that arises from the plasma toroidal rotation velocity (for simplicity, we set $M_{\parallel,tor} = 0$). In this version of the FSM we use the ELM energy fluence instead of the power, and we multiply by an additional factor of two to account for the energy in the full Maxwellian (the original form of this equation from Fundamenski *et al.* (2006) is for a half-Maxwellian (Fundamenski *et al.* 2006)). From this, we can calculate the 'heat-flux factor' (HFF) due to a type-I-ELM by making a semi-infinite slab approximation and convoluting the time-varying ELM heat pulse with the one-dimensional (1-D) conduction system's impulse response (or Green's function) as described in Hu *et al.* (2024)

$$\mathrm{HFF}(t) = \frac{1}{2}\int_0^t \frac{q_{ELM}(\tau)}{\sqrt{t-\tau}}\mathrm{d}\tau. \tag{2.15}$$

As a sanity check, we also calculate the cumulative energy fluence

$$\epsilon_\perp(t) = \int_0^t q_{ELM}(\tau)\mathrm{d}\tau, \tag{2.16}$$

which should approach (2.12) for $t \to \infty$. In figure 3, we show the heat-flux profile, heat-flux factor and cumulative energy fluence calculated for the minimum type-I-ELM energy fluence expected for ARC, corresponding to $C = 1$. We find an instantaneous peak heat flux of $2.4\ \mathrm{GW\ m^{-2}}$ and a peak heat-flux factor of $23\ \mathrm{MJ\ m^{-2}\ s^{-0.5}}$ for a small ($C = 1$) type-I-ELM, and a instantaneous peak heat flux of $7.3\ \mathrm{GW\ m^{-2}}$ and a peak heat-flux factor of $70\ \mathrm{MJ\ m^{-2}\ s^{-0.5}}$ for a large ($C = 3$) type-I-ELM on ARC. This is of the order of the material limit for tungsten PFCs of approximately $50\ \mathrm{MJ\ m^{-2}\ s^{-0.5}}$, indicating that type-I-ELMs will either need to be mitigated or completely avoided in ARC.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

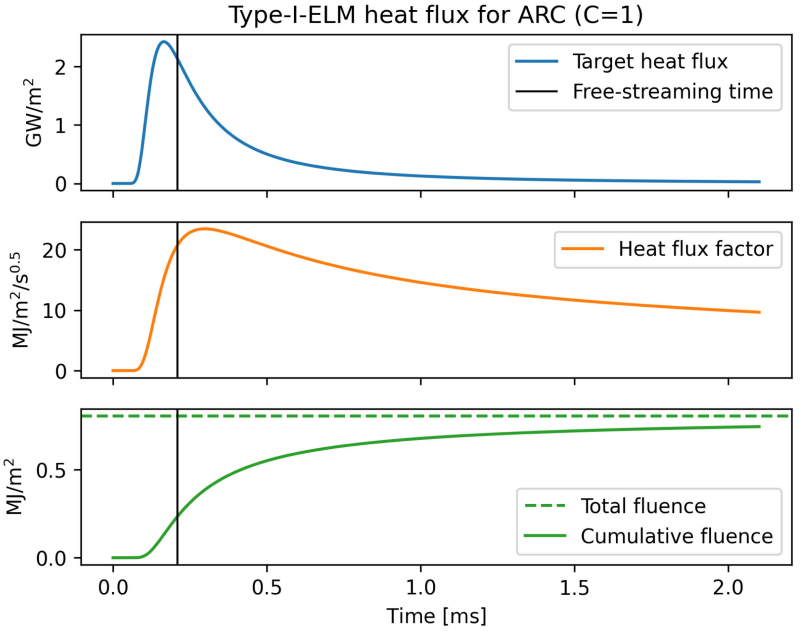


FIGURE 3. Time-varying profiles for the minimum type-I-ELM energy fluence ($C = 1$) predicted for ARC, showing the target heat flux (top, calculated using (2.14)), heat-flux factor (middle, (2.15)) and cumulative energy fluence (bottom, (2.16)).

2.4. *Maintaining detachment during heat-flux transients*

ARC will operate with detached divertors to limit erosion, which may provide some buffering against transient heat fluxes (Henderson *et al.* 2023; Komm *et al.* 2023; Flanagan *et al.* 2025). Significant reductions in the ELM energy fluences can be achieved via impurity seeding (Komm *et al.* 2023), and if we fully detach the divertors and hold the detachment front away from the divertor targets, the energy cost to ionise the neutral particles should further reduce the ELM energy fluence (Henderson *et al.* 2023). We can make a rough estimate of how ELMs will interact with a detachment front by estimating the number of neutrals which will be ionised by an ELM and the perpendicular area over which the ELM will deposit its energy. For a transient with a parallel energy fluence of $\epsilon_\parallel$, we expect that the parallel length of neutrals ionised by the ELM will be

$$L_{iz} = \frac{\epsilon_\parallel}{n_0 \cdot \varepsilon_H \cdot \tau_{resid}/\tau_{iz}}. \tag{2.17}$$

Taking the divertor neutral pressure of 20 Pa calculated in § 4.4, we can calculate the divertor neutral density $n_0 = 2p_{div}/T_{wall} \approx 3 \times 10^{21}$ m^{-3} (where the factor of two converts from molecular to atomic density). For the hydrogen ionisation energy ε_H, we can either (conservatively) use the first ionisation potential $\varepsilon_H = 13.6$ eV or (optimistically) an effective ionisation energy $\varepsilon_H \approx 25$ eV (see § 3.5.2 of Stangeby (2000)). For the ratio of the particle residence time to the characteristic ionisation time in the divertor τ_{resid}/τ_{iz}, we can either (conservatively) use the value of $\tau_{resid}/\tau_{iz} \sim 30$ for ASDEX-Upgrade from Henderson *et al.* (2023) or (optimistically) calculate $\tau_{resid}/\tau_{iz} = (L_\parallel n_{e,t} k_{iz}(T_{e,t})/\sqrt{2T_{e,t}/\langle m_i \rangle}) \sim 4000$ (equation 7 from Henderson *et al.*

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

(2023)) using values calculated for ARC in § 4.4. Using conservative assumptions, we calculate that a type-I-ELM with energy fluence in the range $\epsilon_{\parallel} = 18\,\mathrm{MJ\,m^{-2}}$ to $2.4\,\mathrm{MJ\,m^{-2}}$ ($C = 1 - 3$) will ionise between 90 and 300 m of neutrals, while using optimistic assumptions we find between 0.4 and 1 m will be ionised. Compared with the parallel connection length of the divertor $L_{div} = 73$ m (see § 3), with conservative assumptions $L_{iz} > L_{div}$ which we interpret as implying that ELMs will cause divertor reattachment, while with optimistic assumptions $L_{iz} \ll L_{div}$ implying that the neutrals in the divertor may be able to absorb the energy of the ELM. However, when we also calculate the radial extent of an ELM by calculating the ELM-wetted area, we find that ELMs will deposit their energy much further into the SOL than the continuous heat exhaust. Using the pedestal values from the previous section, we can calculate the pedestal energy $W_{ped} = 3/2 n_{e,ped}(T_{e,ped} + T_{i,ped}) V_{plas} \approx 81$ MJ (for $V_{plas} = 189\,\mathrm{m^3}$ from Hillesheim *et al.* (2026)) and the pedestal top collisionality

$$\nu^*_{e,ped} = \frac{\pi q_{cyl} R}{1.03 \cdot 10^{16}} \frac{n_{e,ped}}{{T_{e,ped}}^2} Z_{eff,ped} \approx 0.075 \tag{2.18}$$

for $n_{e,ped}$ in $\mathrm{m^{-3}}$ and $T_{e,ped}$ in eV, with $Z_{eff,ped} = 1.7$. This lets us estimate $\Delta W_{ELM}/W_{ped} \approx 10\,\%$ using the relationship from figure 11 of Loarte *et al.* (2003), and therefore $\Delta W_{ELM} \approx 8.1$ MJ. The radial extent of an ELM can be calculated using

$$\lambda_{q,ELM} = \frac{\Delta W_{ELM}}{\epsilon_{\perp}} \times \frac{0.5}{2\pi\, R_{axis} f_x}, \tag{2.19}$$

where the factor of 0.5 accounts for both divertor target plates, and $f_x \approx 6$ is the flux expansion which lets us map from the midplane to the divertor target. Taking $\epsilon_{\perp}$ for a large ($C = 3$) ELM from the previous section, we calculate $\lambda_{e,ELM} \approx 10$ mm. This is significantly larger than the inter-ELM heat-flux decay length of approximately 0.56 mm (from table 1), implying that type-I-ELMs on ARC will deposit their energy far into the SOL which is typically more challenging to detach than the near-SOL (Kallenbach *et al.* 2015). Therefore, due to both the parallel and perpendicular dynamics, we view buffering type-I-ELMs as too uncertain to be taken as the power exhaust solution for ARC.

Nevertheless, it may be possible to buffer against smaller heat exhaust transients such as the filaments which are characteristic of the QCE scenario discussed in § 6.1. Recent work by Perillo *et al.* (2026) shows that the energy fluence of small ELMs and filaments can be described by (2.8) by replacing the pedestal top pressure by the pedestal foot pressure. Taking separatrix values computed in § 4, we find that filaments generated near the separatrix should be able to buffered by between 3 and 8 m of neutrals using conservative assumptions, which is significantly less than the parallel connection length of the divertor. Therefore, by operating ARC with a detachment front in the divertor leg, we should be able to buffer against smaller ELMs and filaments – providing protection against both the continuous and transient heat fluxes predicted for ARC.

3. Designing the ARC divertor

To limit erosion ARC will need to operate with detached divertors, and to provide a sufficient buffer against heat-flux transients the detachment front should be held away from the divertor targets. One option would be to allow the detachment front to reach the primary X-point and control the resulting 'X-point radiator' (XPR)

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

(Bernert *et al.* 2025). However, as discussed in § 6.2, until we can test the performance of the XPR scenario on SPARC, we want to ensure that we can stabilise the detachment front in the divertor leg. In § 3.1, we discuss current experimental results and modelling to identify effects which passively stabilise detachment, and in § 3.2 we introduce the X-point target geometry which maximises these effects. We then present the current divertor design for ARC in § 3.3 and discuss ongoing work to optimise the divertor design in § 3.4.

3.1. *Stabilising a detachment front in the divertor leg*

Stabilising a detachment front in the divertor leg is challenging. On tokamaks such as ASDEX Upgrade, DIII-D and JET, once the detachment front lifts off the divertor targets, it moves rapidly to the X-point (Lipschultz, Parra & Hutchinson 2016). Conversely, on MAST-U and TCV it is possible to stabilise the detachment front in the divertor leg (Ravensbergen *et al.* 2021; Verhaegh *et al.* 2025). Understanding the differences between these tokamaks helps to identify effects which should improve detachment stability in ARC. Firstly, both MAST-U and TCV have carbon walls, and typically have less exhaust power than ASDEX Upgrade, DIII-D and JET. If SPARC experiments find that detachment stabilisation is not possible in a high-powered, metal-walled tokamak, we will need to change our power exhaust solution for ARC. However, in addition to these factors, both MAST-U and TCV can operate in advanced/alternative divertor configurations (such as the 'X-divertor' (Kotschenreuther *et al.* 2007), 'super-X' (Valanju *et al.* 2009) and 'X-point target' configurations) and with physical baffles to confine neutrals in the divertor leg (Reimerdes *et al.* 2021; Kool *et al.* 2025; Lee *et al.* 2025; Verhaegh *et al.* 2025).

Modelling (Kotschenreuther *et al.* 2013; Lipschultz *et al.* 2016; Cowley *et al.* 2022; Carpita *et al.* 2024) and experimental results (Lee *et al.* 2025; Verhaegh *et al.* 2025) suggest that tightly baffled, long-legged divertors with significant flux expansion may be able to stabilise a detachment front in the divertor leg. A mechanism for this is sketched in figure 4. If the magnetic field strength decreases in the direction of the divertor targets, $q_{\parallel}$ will decrease and the area for plasma–neutral interactions will increase in the direction of the divertor targets. If a detachment front moves towards the divertor target due to a transient increase in P_{sep} or decrease in $n_{e,u}$ or c_z, the rate of power dissipation will increase, pushing the detachment front back towards its original position. Conversely, if a detachment front moves upstream, these effects should act in the opposite direction. Therefore, with sufficient flux expansion in the divertor leg, we should be able to stabilise a detachment front (LaBombard *et al.* 2015; Verhaegh *et al.* 2025). However, in TCV experiments, this is only observed with tight divertor baffling (Theiler *et al.* 2017). Without divertor baffling, strong parallel particle flows form due to neutral ionisation upstream of the X-point (Fil *et al.* 2020; Cowley, Moulton & Lipschultz 2026). These parallel flows shift heat flux from the conductive to the convective channel and increase the static pressure loss from the outboard midplane to the detachment front – reducing the plasma density in front of the detachment front and making it more difficult to access detachment (Carpita *et al.* 2024; Cowley *et al.* 2024). By introducing baffles, the benefits of total flux expansion are recovered (Fil *et al.* 2020; Lee *et al.* 2025) – suggesting that a combination of total flux expansion and baffling are needed for stabilising a detachment front (Verhaegh *et al.* 2025).

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

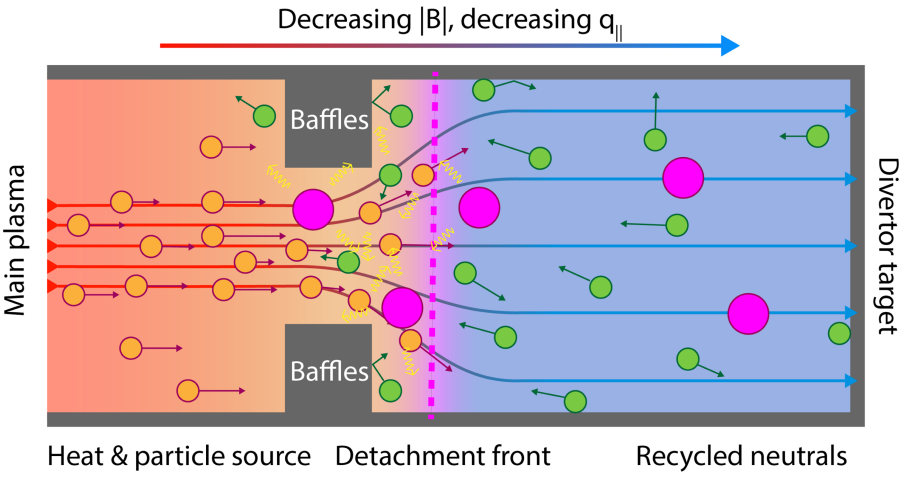


FIGURE 4. The effect of magnetic flux expansion and baffles on detachment stability. Orange circles indicate main ions, green circles indicate neutrals and purple circles indicate a seeded impurity. A gradient in the magnetic field strength will lead to a gradient in $q_{\|}$ due to the changing area of the flux tube, and physical baffles cause recycled neutrals to be primarily ionised in the divertor. These effects can passively stabilise a detachment front in the divertor leg, as well as reduce the impurity concentration required to access detachment.

3.2. *The X-point target configuration*

The 'X-point target' (XPT) advanced divertor geometry aims to maximise the detachment stability of the outer divertor leg by introducing a secondary X-point in the outer divertor leg, close to the outer divertor target (LaBombard *et al.* 2013, 2015).[2] In a ideal XPT configuration, the secondary X-point is positioned close to the separatrix of the primary X-point, ideally in the SOL common flux region on a flux surface between 0.5 and 1 $\lambda_{q,u}$ into the near-SOL (when mapped back to the outboard midplane). This increases the connection length and flux expansion of the flux tube carrying the maximum heat flux, and may also lead to a redistribution of the heat flux over multiple strike points through strike-point splitting. This configuration was originally proposed for the ADX concept (LaBombard *et al.* 2015), for which it was predicted that the XPT would stabilise a detachment front (an 'X-point MARFE') at the secondary X-point, leading to passively stable full detachment. In UEDGE simulations of ADX, the XPT configuration allowed for detachment access up to $\sim 5-10\times$ higher P_{sep} than for a standard vertical plate divertor (Umansky *et al.* 2017). In these simulations the detachment front was found to be passively stable, with the stability window increasing with the divertor leg length (Umansky *et al.* 2020). The XPT was also studied for an earlier design of ARC (Kuang *et al.* 2018), where UEDGE simulations found that the XPT geometry reduced the target electron temperature by a factor of ~ 10 relative to a super-X configuration if the primary and secondary X-points were aligned to within $\sim 1.4\lambda_{q,u}$ (when mapped to the outboard midplane) (Wigram *et al.* 2019). These simulation results have largely been supported by experimental results from TCV, which find that the XPT geometry reduces the outer divertor sheath-entrance temperature and power flux relative to a conventional divertor configuration or even X and super-X configurations (Raj

[2]This is topologically similar to the 'snowflake-minus' (Ryutov 2007; Piras *et al.* 2009 and Soukhanovskii et al. (2011) or 'tripod' configuration Zheng et al. (2014)

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

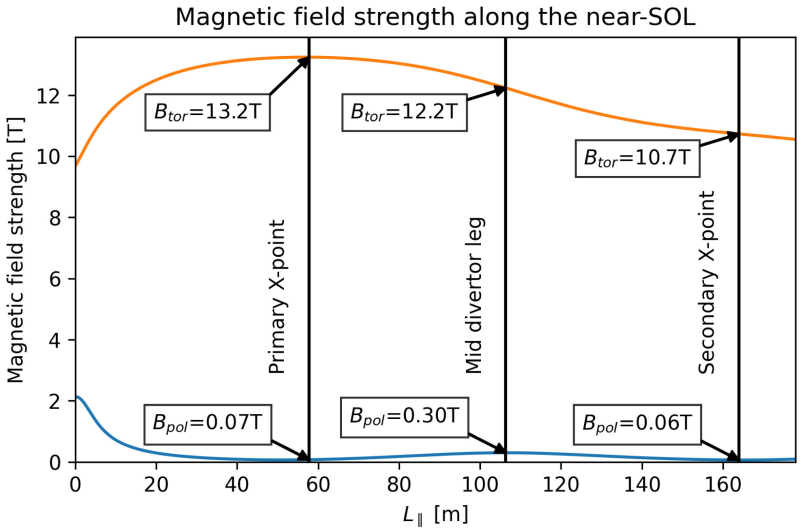


FIGURE 5. Magnetic field along the near-SOL (surface marked in purple in figure 1, 1 mm into the SOL when measured at the outboard midplane), from the outboard midplane to the divertor target. The toroidal and poloidal field strengths are marked in orange and blue respectively, with values given at the primary X-point, in the mid-divertor leg and at the secondary X-point.

et al. 2022; Lee *et al.* 2025). These experiments also confirmed the prediction of LaBombard *et al.* (2015) that a detachment front could be stabilised at the secondary X-point – observing the formation of an 'XPT radiator' similar to an XPR except located at the secondary X-point (Lee *et al.* 2025).

3.3. *The divertor geometry for the ARC V3A design point*

To increase the range of upstream conditions for which we can access detachment and improving the passive stability of the detachment front in the divertor leg, we have developed a X-point-target divertor design for ARC. As shown in figure 1 and discussed in Hillesheim *et al.* (2026), our design features a long, tightly baffled divertor leg, with a secondary X-point in the outer divertor leg. This geometry maximises the total flux expansion from the primary X-point at $R_{xpt} = 3.9$ m to the divertor target at $R_t = 5.1$ m by placing the divertor as far radially outwards as possible. As shown in figure 5, at a flux surface 1 mm from the separatrix (when mapped to the outboard midplane), this configuration has a toroidal flux expansion of $\sim$1.15 and a poloidal flux expansion of $\sim$5.5 between the divertor leg mid-point and the secondary X-point. The secondary X-point also increases the parallel connection length: for the same field line, we find a parallel connection length of 170 m from the outboard midplane to the divertor target, of which 123 m are in the divertor leg. However, for the calculations in this paper, we decided to reduce the divertor leg length to 73 m, reducing the total connection length to 120 m. This reduction is conservative and accounts for possible challenges in achieving an ideal XPT configuration.

3.4. *Ongoing development of the ARC divertor design*

The divertor design will be optimised further in future ARC design iterations. Due to the flexibility afforded by the liquid FLiBe coolant and breeding blanket, the design of the ARC first wall and vacuum vessel can evolve until relatively late in

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

ARC's design and construction. This means that we can use SPARC experiments to finalise the design of ARC's divertor, significantly reducing the extrapolation risk of the final design. We note that the first-wall geometry shown in figure 1 is not yet finalised, and will change in future design iterations. One topic of research for setting the wall geometry is determining the optimal width of the divertor leg, since a narrower leg improves neutral baffling at the cost of increasing the heat flux to the baffles. Different options for the magnetic geometry will also be explored. In particular, since the XPT geometry performs similarly to a super-X divertor (SXD) (Valanju *et al.* 2009) if the secondary X-point is beyond $\sim 1 - 2\lambda_{q,u}$ of the primary separatrix (Wigram *et al.* 2019), we will explore whether a SXD is sufficient for power exhaust on ARC. A SXD configuration would relax the requirements on magnetic control relative to an XPT. The SXD configuration has been shown to improve heat-flux handling and detachment stability on MAST-U (Kool *et al.* 2025), but it is not yet clear whether this is sufficient for ARC's increased heat fluxes and long-lifetime requirements.

These research questions will be the focus of SPARC's 'Advanced Divertor Mission'. This mission will focus on testing integrated divertor scenarios that project well to long pulse operations under power-plant-relevant conditions in SPARC, exploring a variety of magnetic topologies including long-leg designs and advanced topologies like the SXD and XPT (Kuang *et al.* 2020). To test these divertor solutions under ARC-like plasma conditions, SPARC will run experiments approaching ARC's $P_{sep}B_{axis}/(R_{axis}n_{e,u}^2)$ values, SOL collisionality $\nu_{e,edge}^*$ and α_t. These experiments will aim to identify highly dissipative at conditions close to those expected for ARC, focusing on scenarios which can maintain stable detachment at moderate divertor impurity seeding levels, integrated with helium exhaust and a suitable small-ELM scenario. Experimental results from this mission will also provide a critical validation point for high-fidelity SOL models, which – once validated – will be used to further refine the design of ARC's divertor.

4. Detachment-compatible scenarios

The long-legged, tightly baffled XPT divertor design presented in § 3 should increase the range of core-plasma conditions for which we can access detachment, but we still need to check that we can detach a high fusion power scenario for ARC. In § 4.1 we introduce core modelling performed for ARC, focusing on identifying the separatrix density which leads to the highest fusion performance. Then, in §§ 4.2 and 4.3, we introduce the 'extended Lengyel' model, which lets us quantitatively calculate the impurity concentration required to access detachment access. We then apply this model in § 4.4 to calculate detachment onset conditions corresponding to different core scenarios, and show that ARC should be able to access detachment with seeding levels comparable to existing tokamaks.

4.1. *Calculating the fusion performance versus separatrix density*

A workflow to predict the core performance of ARC scenarios was developed in Howard *et al.* (2026). This workflow used a neural network trained on EPED (Snyder *et al.* 2011) simulations of the ARC pedestal over a broad range of conditions, combined with parametrised core profiles inside the pedestal, which allow for the rapid prediction for ARC's fusion power generation. The results of this workflow were verified against both ASTRA and coupled TRANSP-PORTALS predictions, and found to typically agree within ∼10 %. For the results shown in

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

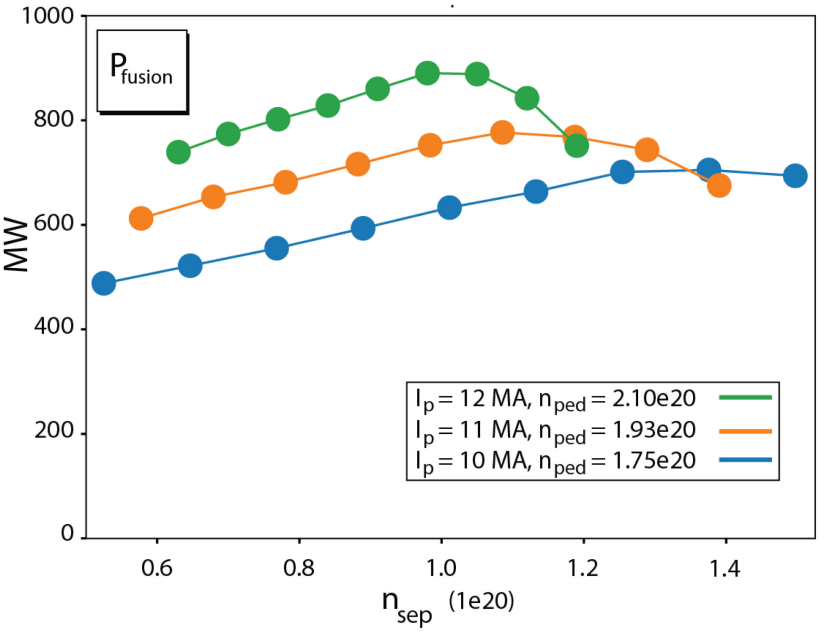


FIGURE 6. ARC fusion power versus $n_{e,u}$, for I_p=10–12 MA, fixed $B_{axis} = 11.4\,\mathrm{T}$, $q^* = 3.2-3.8$ ($q_{cyl} = 3.0-3.6$) and a fixed Greenwald density fraction of 0.9.

figure 6, we used the core-pedestal modelling workflow to calculate how the fusion power output of ARC varies as a function of the separatrix density. For these simulations we used $B_{axis} = 11.4\,\mathrm{T}$ for three different values of I_p, for a flat $Z_{eff} = 1.5$ profile (corresponding to a tungsten concentration of $n_W/n_e = 1.5 \times 10^{-5}$ and a 'lumped impurity' selected to maintain a dilution of $n_{DT}/n_e = 0.85$). For each of the three currents, the fusion power increased with density up to the imposed limit of 90 % of the Greenwald density limit (Greenwald *et al.* 1988). In figure 6, we show results at 90 % of the density limit and at a separatrix power of $P_{sep} = 120\,\mathrm{MW}$, which is approximately $P_{sep}/P_{L-H} \sim 1.2\times$ the LH threshold power predicted at this density (Hillesheim *et al.* 2026). For each plasma current tested, we varied the ratio of $n_{e,u}/\langle n_e \rangle$ at fixed $\langle n_e \rangle$ to investigate the effect of the separatrix density on core-pedestal performance. We find that the predicted fusion power increases with separatrix density when the pedestal is limited by current-driven peeling modes, until the pedestal reaches the ballooning branch where further increases in the separatrix density cause a significant decrease in the fusion power. Because the core turbulent transport is stiff, pinned near a critical gradient scale length, the variation in the predicted pedestal height with separatrix density is the primary driver of these changes in fusion power.

At the highest plasma current of $I_p = 12\,\mathrm{MA}$, the highest fusion power of $P_{fus} \sim 890\,\mathrm{MW}$ was found at $n_{e,u} = 1\times10^{20}/\mathrm{m}^3$. This is below the 1100 MW predicted using empirical POPCONs, but consistent with the range of uncertainty of medium-fidelity modelling (Hillesheim *et al.* 2026; Howard *et al.* 2026). As noted in Hillesheim *et al.* (2026), one path to increasing the fusion power output would be to operate with a Greenwald fraction higher than 90 %, which would have the added benefit of reducing the impurity concentration required to access detachment. Compared with

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

the $I_p = 12$ MA $P_{fus} \sim 890$ MW, at $I_p = 11$ MA the maximum fusion power reduces to 780 MW at $n_{e,u} = 1.1 \times 10^{20} \mathrm{m}^{-3}$ and at $I_p = 10$ MA the maximum fusion power reduces to 700 MW at $n_{e,u} = 1.4 \times 10^{20} \mathrm{m}^{-3}$. Interestingly, the reduction in maximum fusion power from $I_p = 12$ MA to $I_p = 10$ MA is a factor of 1.26, which is less than the factor $1.2^2 = 1.44$ reduction that would be expected due to the reduction in the Greenwald density limit. The shift to higher separatrix density at reduced current will reduce the impurity concentration required to access detachment, and as such operating at slightly reduced I_p may be attractive if the reduction in P_{fus} can be offset by operating at a higher Greenwald fraction.

4.2. *The 'extended Lengyel' model for detachment onset*

To identify a core-edge integrated scenario that maximises the fusion power of ARC, we need a model which can quantitatively predict the impurity concentration needed to access detachment. The impurity concentration links the divertor power exhaust solution to the core performance, since excessive seeding can result in core fuel dilution and increased flux consumption. To calculate it, we developed a simple 'extended Lengyel' model in Body, Kallenbach & Eich (2025), building on the model developed in Lengyel 1981).

The extended Lengyel model calculates the impurity concentration c_z required to decrease the parallel heat flux in the SOL from its upstream value $q_{\parallel,u}$ – given by (2.2) – to a tolerable target value $q_{\parallel,t}$ – which we can relate to target electron temperature and sputtering yield via (2.6). We assume that the static pressure is constant in the SOL and that impurity radiation is the dominant power loss term. Compared with the original Lengyel model, in the extended Lengyel model we added corrections for the pressure and power loss due to recycled neutrals, and added a correction for cross-field transport in the divertor. With these corrections, the extended Lengyel model predicts that the impurity concentration required for detachment is

$$c_z = \frac{\left(1 + \left(1/b_{div}^2 - 1\right)(1 - f_{rad,main})^2 - \left((1 - f_{pow})/(1 - f_{pow,c})\right)^2\right) {q_{\parallel,u}}^2}{2(\kappa_{e,0}/\kappa_z) n_{e,u}^2 T_{e,u}^2 L_{INT}}, \tag{4.1}$$

where $b_{div} = \lambda_{INT}/\lambda_{q,u}$ is a heat-flux broadening factor (discussed in § 4.3), $f_{rad,main}$ is the fraction of power radiated above the divertor entrance, $f_{pow,c}$ is the fraction of power lost due to ionisation of recycled neutrals near the divertor targets, $\kappa_{e,0}$ is the electron heat conductivity constant (used in (2.7)), κ_z is a correction to the heat conductivity due to finite Z_{eff} (Brown & Goldston 2021), $n_{e,u}$ and $T_{e,u}$ are the electron density and temperature at the outboard midplane separatrix, and $L_{INT} \approx \int_0^{T_{e,u}} L_z(T_e)\sqrt{T_e} dT_e$ is a weighted integral of the impurity radiation coefficient L_z (Pütterich *et al.* 2019). Unlike in § 2.1, the upstream parallel heat-flux density is calculated using a turbulence-broadened $\lambda_{q,u}$ expression from Eich *et al.* (2020)

$$\lambda_{q,u} = \frac{0.6 \rho_{s,pol,avg} \left(1 + 2.1 \alpha_t^{1.7}\right)}{(R_{axis} + a_{geo}) B_{pol,u} / \left(R_{axis} \langle B_{pol} \rangle_{sep}\right)}, \tag{4.2}$$

which agrees with the 'heuristic drift model' (Goldston 2012) for $\alpha_t \to 0$ and increases by a factor of ~3 for $\alpha_t \to 1$. Following the method of Kallenbach *et al.* (2016), we also calculate the divertor neutral pressure from the target-perpendicular

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

ion flux $\Gamma_{i,\perp,t}$

$$p_{div} \equiv \frac{\Gamma_{i,\perp,t} T_{wall}}{\sqrt{\frac{T_{wall}}{\pi m_i}}}. \tag{4.3}$$

The derivation of this model and details on how these terms are calculated are available in Body *et al.* (2025). Unlike the original Lengyel model which systematically overestimates the impurity concentration required to access detachment (Moulton *et al.* 2021; Henderson *et al.* 2021), the extended Lengyel model accurately calculated the impurity concentration and divertor neutral pressure at detachment onset of an ASDEX Upgrade experiment (Body *et al.* 2025). The extended Lengyel model also reproduced the empirical scaling from Kallenbach *et al.* (2016), which has been shown to describe detachment access on multiple machines (Henderson *et al.* 2021, 2024*a*,*b*). The model is also much faster than transport modelling, running in approximately ∼100 ms per evaluation, which lets us use this model to predict detachment access for a broad range of core conditions.

4.3. *Heat-flux broadening in the divertor*

The extended Lengyel model shows that the impurity concentration required to access detachment depends on the divertor heat-flux broadening factor

$$b_{div} = \frac{\lambda_{INT}}{\lambda_{q,u}} = 1 + \frac{1.64S}{\lambda_{q,u}}. \tag{4.4}$$

The $\lambda_{q,u}$ decay length is well described by the multi-machine scaling (Eich *et al.* 2013) and models such as the heuristic drift model (Goldston 2012). The power spreading factor S describes cross-field transport in the divertor (Eich *et al.* 2013), and this parameter is less well understood than $\lambda_{q,u}$. Several scalings for S have been developed on individual tokamaks (Scarabosio *et al.* 2015; Sieglin *et al.* 2016; Nille *et al.* 2019), but an accepted multi-machine scaling has not yet been developed, and as such we instead focus on models for b_{div}. In the mostly attached database used for the multi-machine scaling, a $S \approx \lambda_{q,u}/2$ relationship was observed (albeit with significant scatter), corresponding to $b_{div} = 1.8$ (Eich *et al.* 2013). Kallenbach *et al.* (2016), to match the empirical detachment-onset scaling a value of $b_{div} = 3$ was used in the model presented in that work. More recently

$$b_{div} \approx 1.7\,(p_{div}/\,\mathrm{Pa})^{0.38} \tag{4.5}$$

was proposed in Henderson *et al.* 2025, based on a database of SOLPS-ITER simulations for STEP (Henderson *et al.* 2025), which would give $b_{div} \approx 5.4$ for $p_{div} \sim 20$ Pa (from § 4.4). Another recent scaling

$$b_{div} = 1 + 2.3\frac{R_t - R_{xpt}}{a_{geo}} \tag{4.6}$$

was proposed by Brida *et al.* (2025), based on Pfirsch–Schlüter flows, which would give $b_{div} \approx 3.3$ for $R_t = 5.1$ m and $R_{xpt} = 3.9$ m. Both (4.5) and (4.6) suggest that we might expect higher divertor broadening on ARC, and as such we use a slightly increased $b_{div} = 4$ when calculating detachment access for ARC. The impact of varying this assumption is shown in figure 8.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

4.4. *Calculating the impurity concentration required for detachment*

We used the extended Lengyel model to calculate the impurity seeding required to access detachment, for each of the three I_p cases shown in figure 6. In table 2 the detachment onset conditions corresponding to the highest P_{fus} point is given for each I_p, in figure 7 we show the impact of varying $n_{e,u}$ and P_{sep} around the maximum P_{fus} operational point, and in figure 8 we show the impact of varying the divertor broadening factor b_{div}. At the 12 MA, 890 MW operational point, we reach detachment with a divertor argon concentration of 0.9 % with a divertor neutral pressure of 20 Pa. The argon concentration is similar to the argon concentration required to access detachment in ASDEX Upgrade and JET (Kallenbach *et al.* 2015); Henderson *et al.* 2024*a*,*b*), since the increased P_{sep} is offset by the increased $n_{e,u}$. This result is in line with the scaling proposed in Reinke (2017), which predicted that although the upstream heat-flux density $q_{\parallel,u}$ will increase in high-field tokamaks (relative to existing tokamaks), the increase in the separatrix density $n_{e,u}$ means that detachment can be achieved without a significant increase in the divertor impurity concentration c_z. The divertor neutral pressure of 20 Pa is somewhat higher than on existing tokamaks, which will have the benefit of increasing the impurity exhaust rate (discussed further in § 5). This is due to the high upstream pressure, which increases the target particle flux $\Gamma_{i,\perp,t} \propto (n_{e,u}T_{e,u})^2$ (according to the two-point model (Stangeby 2018)) and therefore the divertor neutral pressure p_{div} (according to (4.3)). Therefore, while ARC is predicted to have challenging upstream heat fluxes, it is predicted to access conditions with favourable detachment access and impurity exhaust. A caveat to this favourable result is that the α_t turbulence predicted for the 12 MA case is $\alpha_t \sim 0.21$. As discussed in § 2, low values of α_t on existing tokamaks typically correlate with conditions which are more challenging to find a power exhaust solution for – and, in particular, are typically associated with type I ELMy H-modes (Faitsch *et al.* 2023, 2025). Interestingly, as shown in both figure 7, the value of α_t at detachment onset was not strongly affected by varying either $n_{e,u}$ or P_{sep}. To understand why this is the case, we approximated $L_{INT} \propto T_{e,u}$ (from Reinke (2017)) and assumed $q_{\parallel,t} \ll q_{\parallel,u}$ and $L_\parallel \approx \pi q_{cyl} R$. Using these approximations, we can rewrite the basic Lengyel model ((4.1) with $b_{div} \to 1$ and $q_{\parallel,cc} \to q_{\parallel,t} \to 0$) using (2.7) and (2.1) to find

$$c_z \propto \frac{q_{\parallel,u}^2}{n_{e,u}^2 T_{e,u}^3} \propto \frac{T_{e,u}^4}{n_{e,u}^2 q_{cyl}^2} \propto \frac{q_{cyl}^2 Z_{eff,u}^2}{\alpha_t^2}, \tag{4.7}$$

$$\alpha_t \propto \frac{q_{cyl} Z_{eff,u}}{\sqrt{c_z}} \approx q_{cyl}\sqrt{c_z}\ . \tag{4.8}$$

(The full derivation of this result including the proportionality factor is given in Appendix B.) When expressed in this form, α_t at the onset of detachment does not explicitly depend on $n_{e,u}$ or P_{sep}, and only depends on them via their impact on c_z which in turn only weakly affects α_t. According to (4.8), to increase α_t our strongest lever would be to increase the cylindrical safety factor q_{cyl} by reducing the plasma current I_p. At shown in figure 7 and in table 2, reducing I_p from 12 to 10 MA increases α_t by more than the change in q_{cyl}, since c_z also decreases significantly. In addition to increasing α_t and reducing c_z, p_{div} also slightly increases at reduced I_p, such that ARC's power and particle exhaust solutions both become more favourable at reduced I_p. However, as shown in figure 6, these improvements come at the cost of reducing the overall fusion performance (for fixed Z_{eff} and n_{DT}/n_e).

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

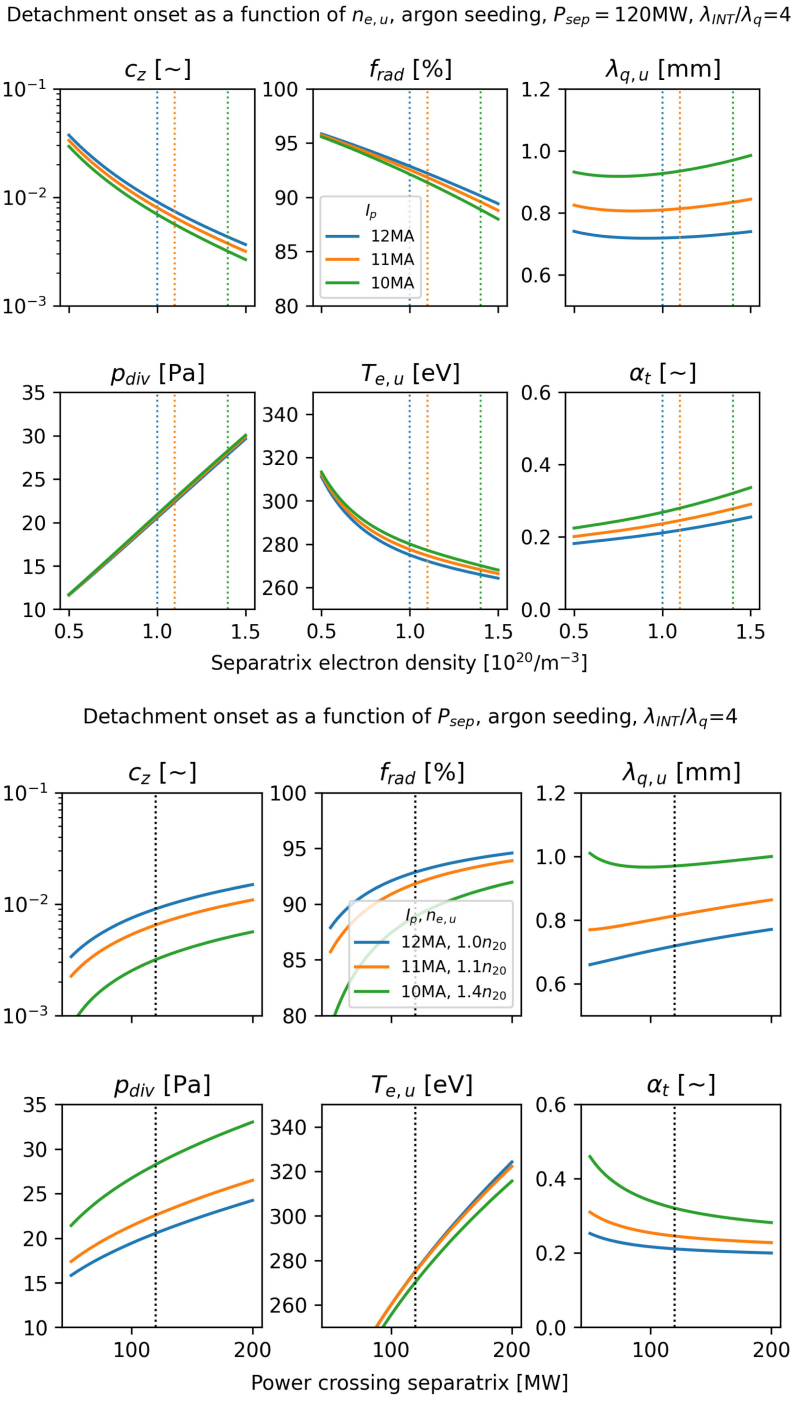


FIGURE 7. Detachment onset conditions with argon seeding as a function of $n_{e,u}$ for a fixed value of $P_{sep} = 120$ MW (upper plots) and as a function of P_{sep} for $n_{e,u}$ giving the maximum P_{fus} in figure 6 (lower plots). Each set of subplots shows (from left to right, top to bottom) the impurity concentration required for detachment c_z, the fraction of power dissipated via radiation f_{rad}, the heat-flux decay length $\lambda_{q,u}$ computed by (4.2), the divertor neutral pressure p_{div}, the separatrix electron temperature $T_{e,u}$ and the computed value of α_t.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

I_p (MA)	12	11	10
q^*	3.2	3.5	3.8
q_{cyl}	2.7	2.9	3.2
$n_{e,u}$ (10^{20} m^{-3})	1	1.1	1.4
P_{sep} (MW)	120	120	120
c_z [%]	0.9	0.65	0.32
f_{rad} [%]	93	92	89
$\lambda_{q,u}$ [mm]	0.72	0.81	0.97
p_{div} [Pa]	21	23	28
$T_{e,u}$ [eV]	275	275	270
α_t	0.21	0.25	0.32

TABLE 2. Detachment onset conditions for maximising P_{fus}, for the three cases shown in figure 6, giving the plasma current (I_p), two approximations for the edge safety factor (q^* and q_{cyl}), the upstream density at the maximum P_{fus} point ($n_{e,u}$), the power crossing the separatrix P_{sep}, the argon concentration required to access detachment c_z, the fraction of power radiated in the SOL f_{rad}, the upstream heat-flux width $\lambda_{q,u}$, the divertor neutral pressure p_{div}, the upstream temperature $T_{e,u}$ and the α_t turbulence parameter.

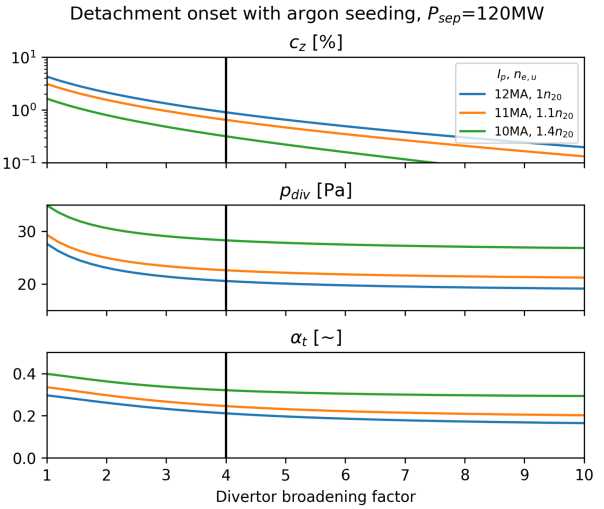


FIGURE 8. Detachment access conditions with argon seeding, predicted by the extended Lengyel model, at a fixed value of $P_{sep} = 120$ MW for $n_{e,u}$ giving the maximum P_{fus} in figure 6, for a scan of divertor broadening values (b_{div}) for three values of q^*.

5. Particle exhaust and enrichment

In addition to managing the heat exhausted from the core, the divertor must also pump impurities out of the plasma. As discussed in § 4, we need to inject impurities into the plasma to access detachment, and we will need to prevent these impurities from polluting the core plasma. In addition, ARC will continuously produce helium 'ash' due to fusion reactions in the core, and this ash must be continuously pumped out of the plasma prevent excessive fuel dilution. In §§ 5.1 and 5.2 we calculate the enrichment required for seeded argon and for helium ash respectively. We find that

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

we should be able to limit the dilution and $Z_{eff,core}$ with reasonable enrichment values, comparable to values observed on ASDEX Upgrade. However, we also note that there is significant uncertainty in this result, and in § 5.3 we discuss how SPARC experiments will be used to study and identify how to improve enrichment on ARC.

5.1. *The impact of divertor impurity seeding on the core*

To access detachment for the $I_p = 12$ MA case given in § 4.4, we need a divertor argon concentration of $c_z = 0.9\,\%$. To determine whether this is compatible with the core scenario given in § 4.1, we need to know the ratio of the core impurity concentration to the divertor impurity concentration. If we assume that the concentration of neutral impurities relative to neutral fuel atoms in the divertor matches the concentration of impurity ions relative to fuel ions, we can assume that this ratio is approximately equal to the impurity enrichment

$$\eta_{Ar} = \frac{n_{0,Ar,div}/n_{0,fuel,div}}{n_{Ar,core}/n_{e,core}} \approx c_{Ar,div}/c_{Ar,core}. \tag{5.1}$$

Using the modelling introduced in § 4.1, we find that the core can tolerate up to approximately $Z_{eff,core} \sim 2$ before performance starts to degrade. Since this workflow already assumes $Z_{eff,core} = 1.5$ due to intrinsic impurities, to avoid performance degradation due to seeding we will need to limit $\Delta Z_{eff,core}$ due to argon to be < 0.5. At the ~10 keV expected in the core, argon ions will be close to fully stripped with a mean charge of $\langle Z \rangle \sim 18$. Therefore, to limit $\Delta Z_{eff,core} < 0.5$ we need to limit the core argon concentration of $\lesssim 0.16\,\%$, corresponding to an argon enrichment of $c_{Ar,div}/c_{Ar,core} \sim 0.9\ \%/0.16\,\% \sim 6$ and a core fuel dilution of $\Delta(n_{DT}/ne) = -3\,\%$. We can compare this with an empirical model for enrichment from Kallenbach *et al.* (2024)

$$\eta_z = 41 \cdot Z^{-0.5} \cdot (p_{div}/\mathrm{Pa})^{-0.4} \cdot \left(E_{iz,z}/E_{iz,fuel}\right)^{-5.8}, \tag{5.2}$$

$$\eta_{Ar} \approx 41 \cdot 18^{-0.5} \cdot 20\ \mathrm{Pa}^{-0.4} \cdot (15.76\ \mathrm{eV}/13.6\ \mathrm{eV})^{-5.8} = 1\,. \tag{5.3}$$

Due to ARC's high divertor neutral pressure of $p_{div} = 20$ Pa, (5.2) gives an enrichment significantly lower than the $c_{Ar,div}/c_{Ar,core} \sim 6$ we need to limit $\Delta Z_{eff,core}$ to < 0.5. However, due to ARC's tightly baffled divertor leg and higher temperatures, we expect that it should achieve higher enrichment than ASDEX Upgrade. In § 5.3 we discuss how SPARC experiments could identify mechanisms for increasing enrichment.

5.2. *Helium exhaust*

ARC will also need to limit the concentration of helium ash in the core plasma to avoid excessive fuel dilution (Reiter *et al.* 1991). Several tokamaks have found that helium exhaust is limited by pumping and not by core transport (Wade *et al.* 1995; Bosch *et al.* 1997; Sakasai *et al.* 1999), and as such the divertor plays a key role in achieving sufficient helium exhaust. The physics of helium exhaust is complex and challenging to accurately model, even with multi-physics transport models such as SOLPS-ITER (Masline *et al.* 2025; Zito *et al.* 2025). Transport simulations and SPARC experiments will be used to optimise and improve ARC's helium exhaust, but until those results are available we focus on providing a simple order-of-magnitude estimate to determine whether particle exhaust is feasible for ARC. For

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

a $P_{fus} = 1100$ MW operational point, the helium production rate will be

$$\frac{dN_{He}}{dt} = \frac{P_{fus}}{\varepsilon_{DT}} = \frac{1100\,\text{MW}}{17.6\,\text{MeV}} = 3.2 \times 10^{20}\,\text{s}^{-1}. \tag{5.4}$$

In steady state, the helium production rate must be balanced by the helium pumping rate

$$\frac{dN_{He}}{dt} = \frac{p_{div} \cdot S_{pump} \cdot c_{He,div}}{T_{He}} = \frac{p_{div} \cdot S_{pump} \cdot \eta_{He} \cdot c_{He,core}}{T_{He}}. \tag{5.5}$$

If we assume that by the time helium is pumped it has thermalised with the walls such that $T_{He} \sim T_{wall} \approx 900$ K, we can calculate the helium enrichment required to limit the core helium concentration below a given $c_{He,core,max}$

$$c_{He,div}/c_{He,core} = \frac{P_{fus}}{\varepsilon_{DT}} \frac{T_{wall}}{p_{div}\ S_{pump}\ c_{He,core,max}}. \tag{5.6}$$

Using $p_{div} \sim 20$ Pa from § 4.4 and making the assumption that ARC will achieve a similar pumping rate to ASDEX Upgrade of $S_{pump} \sim 30\,\text{m}^3/\text{s}$ (Kallenbach *et al.* 2024), to keep the core helium concentration below $c_{He,core,max} < 2\,\%$ we need an enrichment of $c_{He,div}/c_{He,core} \sim 0.4$. This is not significantly more than the $c_{He,div}/c_{He,core} \sim 0.3$ from (5.2), suggesting that ARC should be able to limit the core helium concentration to a few per cent. However, this result is highly uncertain, and ARC's helium exhaust solution will be investigated in future transport simulations and in SPARC experiments.

5.3. *Impurity enrichment*

To integrate detachment with a high-performance core scenario and prevent helium ash accumulation in the core, ARC will need to achieve sufficient impurity enrichment levels. Although the argon and helium enrichment levels calculated in §§ 5.1 and 5.2 are within the range of what has been observed on existing tokamaks, ARC's higher divertor neutral pressure leads to reduced enrichment according to (5.2). With reduced argon enrichment, integrating a high-performance core with divertor detachment is challenging, since impurity seeding in the divertor corresponds to a significant increase in core fuel dilution and a reduction in the pedestal pressure due to increased Z_{eff}. Similarly, reduced helium enrichment increases the pumping rate needed to limit core dilution due to helium ash, resulting in significant additional cost and complexity for the ARC plant. Understanding and optimising impurity enrichment would make finding heat and particle exhaust solutions for ARC significantly easier. Unfortunately, this is easier said than done: matching experimental results for impurity transport in transport models such as SOLPS-ITER is challenging (Masline *et al.* 2025; Zito *et al.* 2025), and recovering the spatial distribution of impurities is challenging in experiments (Kallenbach *et al.* 2024). Of the measurements and modelling which is available, these find that argon and helium enrichment typically decrease during detachment (Hitzler *et al.* 2020; Zito *et al.* 2023; Kallenbach *et al.* 2024), which may result in reduced enrichment on ARC. Conversely, ARC's tightly baffled long-legged divertor may increase helium compression and pumping (Goetz *et al.* 2001; Zito *et al.* 2023), while the increased SOL and pedestal pressures on ARC should improve impurity screening by reducing the ionisation mean free path for impurities.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

In the SOL, parallel impurity transport is mainly set by the force balance of the frictional coupling with the main ions (typically directed towards the target) and the thermal force from temperature gradients (typically directed away from the target) (Senichenkov *et al.* 2019). To increase impurity enrichment, techniques such as 'pump-and-puff' increase main ion flows in the SOL by increasing both the seeding and pumping rate (Osawa *et al.* 2024), overcoming the thermal force and keeping impurities close to the divertor target (Wade *et al.* 1995, 1998; West *et al.* 1999; Masline *et al.* 2025). However, as noted in § 3.1, strong parallel flows in the divertor can reduce how effectively the detachment front can be stabilised by flux expansion (Carpita *et al.* 2024), suggesting that there is a limit to how much we can improve enrichment using 'pump-and-puff' before it starts to impact detachment stability. Optimisation of target plate geometry and divertor chamber shape could further increase enrichment, by optimising the pattern of neutral recycling and ionisation (Zito *et al.* 2025).

The enrichment could be further improved by designing the plant with some form of direct internal recycling, which would increase the impurity concentration in the exhaust gas and ease the requirements on the pumping system. This could be realised by implementing bypass systems in the pumping plenum; an analysis of the conventional divertor for DEMO suggests a bypass loop from the pumping plenum to the SOL can enhance helium recirculation and increase the helium fraction by exploiting differences between the ionisation behaviour of helium and the hydrogenic species (Igitkhanov, Day & Varoutis 2018). Selective screening techniques for impurities entering the pumping plenum have also been investigated (Field *et al.* 2021), but membrane erosion in the pumping port remains a significant challenge.

6. High-performance, small-ELM scenarios for ARC

In §§ 2.3 and 2.4, we showed that, to protect the divertor from transient heat loading and to limit erosion during reattachment, ARC will need to operate in a scenario which avoids large type-I ELMs. In addition, as shown in §§ 4 and 5, ARC's operational scenario must also maximise core performance, be compatible with detachment, and achieve appreciable rates of impurity exhaust. SPARC experiments will be used to test several candidate scenarios, to identify which provides the best integrated solution for ARC. In this section, we highlight the QCE and XPR scenarios in §§ 6.1 and 6.2 respectively, and discuss the use of RMPs for ELM suppression in § 6.3.

6.1. *The quasi-continuous exhaust regime*

The QCE scenario is a type-I-ELM-free H-mode which is found at high plasma shaping (high elongation and triangularity) and at high separatrix densities. This scenario is characterised by high-frequency filaments which are driven by ballooning modes close to the separatrix, which cause additional transport preventing the pedestal from reaching the peeling-ballooning limit (Harrer *et al.* 2022; Dunne *et al.* 2024; Faitsch *et al.* 2025). Despite this additional transport, QCE plasmas can operate close to the global peeling-ballooning threshold (Dunne *et al.* 2024), achieving energy confinement close to that of a type-I-ELMy H-mode, with $H_{98} \approx 0.85 - 0.95$ found on both ASDEX Upgrade and JET (Faitsch *et al.* 2025). In addition to achieving good core performance, QCE plasmas avoid core impurity accumulation since particle transport through the pedestal is maintained (Faitsch *et al.* 2023; Fenstermacher *et al.* 2025). The scenario has been successfully accessed on several

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

tokamaks (Faitsch *et al.* 2025) including in deuterium-tritium operations in JET (Kappatou *et al.* 2025; Sun *et al.* 2025).

To access the QCE regime, two access criteria were recently proposed by Dunne *et al.* 2024). In this work, the IPED model was used to predict the profile and peeling-ballooning stability of QCE pedestals. This was compared with the stability of ballooning modes at the separatrix, which was computed using heuristic models for pressure gradients near the separatrix (Eich *et al.* 2020; Faitsch *et al.* 2023). By varying the plasma shape and pedestal profiles, Dunne *et al.* (2024) identified shaping and edge-pressure-gradient conditions where local separatrix ballooning modes would be triggered before the onset of a global peeling-ballooning ELM crash. The shaping condition

$$S_d = \kappa^{2.2}(1+\delta)^{0.9} > 3 \tag{6.1}$$

is fulfilled for ARC with

$$S_d = 1.55^{2.2}(1+0.36)^{0.9} = 3.5, \tag{6.2}$$

which approaches the $S_d = 4$ typical for QCE plasmas on ASDEX Upgrade (Dunne *et al.* 2024). The edge-pressure-gradient criterion is given in Faitsch *et al.* (2025) as

$$\alpha_{MHD} = \frac{4\mu_0 R_{axis} q_{cyl}^2 n_{e,u} T_{e,u}}{\langle \lambda_{p_e,sep} \rangle B_{axis}^2} > \alpha_{crit,edge} = 0.64\kappa^{2.2}(1+\delta)^{0.9}. \tag{6.3}$$

To determine the separatrix pressure gradients, we use an empirical model fit to experimental results from highly shaped QCE plasmas on ASDEX Upgrade from Faitsch *et al.* (2023)

$$\langle \lambda_{p_e,sep} \rangle = \rho_{s,pol} \times 1.3(1 + 0.002 {\nu^*_{e,edge}}^2) \tag{6.4}$$

$$= \frac{\sqrt{m_i T_{e,u}}}{e\langle B_{pol,sep} \rangle} \times 1.3 \left(1 + 0.002 \left(\frac{100\alpha_t}{q_{cyl}}\right)^2\right). \tag{6.5}$$

Taking values for the $I_p = 12$ MA case from § 4.4 with $\langle B_{pol,sep} \rangle \sim 3/4\, B_{pol,OMP} = 1.5$ T, we find $\alpha_{MHD} = 2.1$ and $\alpha_{crit,edge} = 2.1$, marginally fulfilling (6.3). Therefore, the $I_p = 12$ MA operational point identified in § 4.1 marginally meets both the shaping and separatrix pressure-gradient requirements, suggesting that ARC should be able to access the QCE regime. However, as discussed in § 4.4, the $I_p = 12$ MA operational point has a low $\alpha_t = 0.21$. This is below an empirical threshold value of $\alpha_t > 0.55$ which was found to describe the transition from a type-I-ELMy regime to the QCE regime on ASDEX Upgrade (Faitsch *et al.* 2023). As discussed in Appendix A, when comparing with values computed by the extended Lengyel model, a threshold of $\alpha_t > 0.43$ may be more appropriate, but this is still above the values calculated in § 4.4. As discussed in § 4.4 and Appendix B, to increase α_t at detachment onset it may be necessary to reduce the plasma current. While this leads to more favourable power and particle exhaust conditions, reducing the plasma current will likely reduce the core performance. Another challenge is that, although QCE plasmas avoid type-I-ELMs, the higher-frequency QCE filaments still carry a significant amount of energy. Although in § 2.4 we found that QCE filaments should be buffered by the detachment front, this has not yet been observed on ASDEX Upgrade (Faitsch *et al.* 2023) or other devices. Outside of the divertor,

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

QCE filaments are also expected to increase transport into the far-SOL and to the main-chamber limiters (Redl *et al.* 2023; Perillo *et al.* 2025; Sun *et al.* 2025). This may lead to increased sputtering from the first wall, which would in turn would cause core contamination (Dux *et al.* 2009) and erosion of the main chamber limiters.

SPARC experiments will provide a crucial test of the access criteria and performance of the QCE regime under conditions relevant for ARC. Experimental data from SPARC should help to confirm the access criteria for the QCE regime, including exploring the effect of plasma elongation and triangularity approaching a double null, and the effect of impurity seeding (which – although weaker than the impact of q_{cyl} – should affect α_t via (4.8)). SPARC will also be able to test whether QCE filaments can be buffered in a long-legged, X-point-target configuration, and will explore whether the first-wall fluxes can be kept at a reasonable level while maintaining a small outer gap for Ion Cyclotron Resonance Heating (ICRH).

6.2. *The X-point radiator regime*

Another attractive candidate scenario for avoiding type-I-ELMs is a sufficiently developed XPR regime (Bernert *et al.* 2020, 2023, 2025). Although the divertor design developed in § 3 was optimised for preventing a detachment front from reaching the primary X-point, it is also possible to feedback-control the size of a detachment front at the primary X-point. For a sufficiently large XPR, it is possible to radiate most of the power inside of the confined region and suppress type-I-ELMs while still achieving $H_{98} \sim 0.7 - 1$ (Bernert *et al.* 2025). The XPR regimes have been accessed and controlled on several existing tokamaks, including ASDEX Upgrade, JET, TCV, KSTAR and WEST (Bernert *et al.* 2025), including in deuterium–tritium fuelled plasmas in JET (Kappatou *et al.* 2025; Sun *et al.* 2025). The XPR can maintain detachment through the ramp-up and ramp-down as well as during the L–H and H–L transitions (Bernert *et al.* 2025), protecting the divertor targets for the entire pulse. Using an XPR could also allow for a simpler divertor design, letting us remove or shorten the long-legged divertor and avoid the need to control the position of secondary X-points.

However, it is not yet clear if ARC would achieve sufficient energy confinement and particle exhaust if operated in an ELM-suppressed XPR regime. The energy confinement in ELM-suppressed XPR plasmas has been observed to decrease by as much as ∼30 % relative to the IPB98(y,2) scaling (Bernert *et al.* 2025; Kappatou *et al.* 2025), which would significantly reduce the fusion power generation of ARC. Another challenge is the high impurity concentrations which are required to access the XPR regime (of the order of ∼10 % at the X-point), and the fact that these impurity concentrations must be located inside of the separatrix. Although an impurity enrichment of ∼5 is observed between the X-point and core in current experiments, explaining and quantitatively predicting enrichment remains challenging (see § 5.3), and as such it is difficult to predict whether the core dilution and $Z_{eff,core}$ in an XPR would be tolerable. Another uncertainty is whether the divertor neutral pressure and helium enrichment is sufficient to achieve sufficient helium pumping in this regime. SPARC experiments will help to reduce these uncertainties, to identify whether an XPR is a viable heat exhaust solution for ARC.

6.3. *Suppressing ELMs using resonant magnetic perturbations*

Another solution would be to operate ARC in a standard H-mode, using resonant magnetic perturbations (RMPs) to suppress type-I-ELMs. By applying a perturbative

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

3-D field of the order of ∼0.1 % of the background field using external 3-D coils, the pedestal pressure profile can be prevented from reaching the peeling-ballooning limit. The exact mechanism for this is still debated (see § 4.1.7 by Kirk *et al* (2025) in Fenstermacher *et al.* (2025)). In the context of the EPED model, it was shown that 3-D field penetration near the top of the pedestal can potentially prevent onset of peeling-ballooning mode instability and reproduce the observed q dependence (Snyder *et al.* 2012; Wade *et al.* 2015; Hu *et al.* 2020). The impact of the 3-D field penetration appears to be due to a combination of parallel transport through induced magnetic islands (Logan *et al.* 2025) and due to enhanced cross-field turbulent transport (Paz-Soldan *et al.* 2024). Plasmas with RMP ELM-suppression typically achieve energy confinement of the order of the IPB98(y,2) scaling (Paz-Soldan *et al.* 2024), and the impurity residence time in such plasmas is typically lower than in type-I-ELMy H-modes (Victor *et al.* 2020), helping to avoid impurity accumulation. The pedestal pressure in RMP ELM-suppressed discharges is typically somewhat below the value predicted by EPED for type I ELM and quiescent H-mode discharges, but in optimal circumstances this reduction is only ∼10 %−15 % (Snyder *et al.* 2012; Wade *et al.* 2015; Fenstermacher *et al.* 2025). The ELM suppression via RMPs has been demonstrated on several devices including DIII-D, ASDEX Upgrade, KSTAR and EAST (Paz-Soldan *et al.* 2024) – while on JET RMPs were found to reduce the magnitude of ELMs, but not completely suppress them (Liang *et al.* 2007; Paz-Soldan *et al.* 2024). However, the exact mechanism for ELM suppression depends on the non-linear plasma response to the perturbing field, which makes it difficult to predict the required field strength or access conditions (Fenstermacher *et al.* 2025; Logan *et al.* 2025). Additionally, integrating RMP ELM-suppression with detachment is challenging due to a combination of 'density pump-out' (Krieger *et al.* 2025), an increase in the LH power threshold (Fenstermacher *et al.* 2025), toroidal peaking of the heat-flux footprint (Scotto d'Abusco *et al.* 2025) and limits on the pedestal density and collisionality (Paz-Soldan *et al.* 2024; Fenstermacher *et al.* 2025). Of these, an pedestal density upper limit of $n_{e,ped} \lesssim 3 \times 10^{19}\,\mathrm{m}^{-3}$ observed across several machines and for a range of engineering parameters and edge collisionalities is particularly concerning (Paz-Soldan *et al.* 2024) – since detaching ARC at this density would require divertor argon concentrations of the order of tens of per cent. SPARC experiments will provide a key test of the conditions required to suppress ELMs with RMPs for conditions approaching those expected for ARC, as well as determining whether this solution is compatible with detachment.

7. Conclusions

To support ARC's mission of reliably providing fusion electricity, we have developed a power exhaust solution capable of protecting the divertor targets from the SOL heat fluxes predicted for ARC. We find that to prevent target erosion, ARC will need to operate with a fully detached divertor. To make it easier to access and stabilise detachment in ARC, we developed a long-legged, tightly baffled divertor design for ARC. This design introduces a secondary X-point in the divertor leg in an XPT configuration, which has been both predicted and shown to improve the stability of detachment fronts to variations in the separatrix power or density.

With this divertor design, we predict that ARC will be able access detachment while achieving its target fusion power output with a reasonable divertor argon concentration around 0.9 %. We find that this level of impurity seeding should not significantly degrade core fusion performance if ARC can achieve an argon enrichment of at least $c_{Ar,div}/c_{Ar,core} = 6$. This is higher than predicted by an empirical

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

regression from ASDEX Upgrade data, but within the range of results reported in Kallenbach *et al.* (2024). We also expect that this scenario will efficiently pump helium due to the high predicted divertor neutral pressure of $p_{div} \simeq 20$ Pa. Even with a helium enrichment of $c_{He,div}/c_{He,core} = 0.4$, we predict that the core helium concentration should stay below a few per cent for pumping rates seen on existing tokamaks.

Due to the long divertor leg and high neutral density in the divertor, we predict that the detachment front should be able to prevent reattachment due to filaments produced close to the separatrix, such as those typical of the QCE regime. However, based on a calculation of the projected type-I-ELM energy fluence, we expect that type-I ELMs will re-attach to the divertor target. Depending on the size of the ELM, this may contribute to erosion of the targets or may even exceed the material limits of the divertor targets. As such, to maximise the lifetime of ARC's divertor, we will need to operate ARC in a scenario that avoids type-I ELMs. Several type-I-ELM-free scenarios are being considered for ARC. We show that for the $\sim$1 GW operational point, the shaping and separatrix conditions are such that this point may operate in the QCE regime, which should allow ARC to achieve good energy confinement while operating with filaments which can be buffered via detachment. For the QCE regime and other type-I-ELM-free scenarios, SPARC experiments in support of ARC will be used to determine the access conditions, performance and compatibility with detachment. SPARC will also be used to help retire risks in the ARC design, such as determining the impurity enrichment achievable in a tightly baffled divertor and testing whether the XPT geometry can stabilise a detachment front under high heat and particle loads. Although SPARC experiments and higher-fidelity modelling are needed to reduce uncertainty, on the basis of our current analysis we find that ARC's divertor design is a credible solution to the power exhaust challenges expected for a fusion power plant.

Acknowledgements

Editor Troy Carter thanks the referees for their advice in evaluating this article.

Funding

This work was funded by Commonwealth Fusion Systems. This material is based upon work supported by the U.S. Department of Energy, Office of Science, Fusion Energy Sciences, under the Milestone-Based Fusion Development Program under Award Number DE-SC0024885.

Disclaimer

This report was prepared as an account of work sponsored by an agency of the United States Government. Neither the United States Government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of the information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

Declaration of interests

All relevant funding sources and author affiliations have been disclosed. The authors report no conflicts of interest.

Data availability statement

The data that support the findings of this study are openly available in Zenodo at http://doi.org/10.5281/zenodo.18869845.

Appendix A. Reduced α_t in the extended Lengyel model

In the extended Lengyel model, we have assume all of the power enters the SOL at the outboard midplane and is carried by electron heat conduction, leading to

$$T_{e,u} \approx \left(\frac{7}{2}\frac{q_{\parallel,u}L_\parallel}{\kappa_{e,0}}\right)^{2/7} \quad \text{(see (2.7))}. \tag{A1}$$

Conversely, in the separatrix operational space (SepOS) the power is assumed to enter the SOL along its entire length and 10 % is assumed to be carried by convection, leading to

$$T_{e,u} \approx \left(\frac{7}{4}\frac{(1-0.1)q_{\parallel,u}L_\parallel}{\kappa_{e,0}}\right)^{2/7} \tag{A2}$$

(see (9) from Eich, Manz & the ASDEX Upgrade Team (2021)). Therefore, for the same $q_{\parallel,u}$, the extended Lengyel model will predict a $T_{e,u}$ value which is $\sim(0.9\times 2)^{2/7}\sim 1.2\times$ higher than in SepOS. This in turn slightly changes $n_{e,u}$, since $T_{e,u}$ is used to identify the separatrix position in the edge Thomson data. If we assume $\lambda_n/\lambda_{T_e}\sim 2$, then $n_{e,u}$ will change by about half as much as $T_{e,u}$. Therefore, α_t calculated by the extended Lengyel model will be

$$n_{e,u}T_{e,u}^{-2}\sim 1.1/1.2^2\sim 0.78 \tag{A3}$$

the value calculated by SepOS. As such, we should be able to access the QCE regime with an α_t of $0.55\times 0.78\sim 0.43$.

Appendix B. Impurity concentration required for detachment as a function of the turbulence parameter α_t

As mentioned in § 4.4, we can express the impurity concentration required for detachment in terms of the turbulence parameter α_t. In (4.8) a proportionality is given, while in this appendix we provide the full derivation including the proportionality factor. Starting from the basic Lengyel model equation for c_z and defining $\lambda_{q,u}$ either from the multi-machine scaling (Eich *et al*. 2013) or the heuristic drift model (Goldston 2012), previous work (Goldston et al. (2017) and Reinke (2017)) finds that

$$c_z \propto \frac{P_{sep}}{B_p\, f_{GW}^2} \propto \frac{P_{sep}\, B_p}{{n_{e,u}}^2\, a^2}. \tag{B1}$$

Using the definitions in Goldston *et al*. (2017) (in particular $\ell_\parallel^* = L_\parallel/(\pi q_{cyl}R)$), units of (W, T, kg, m, eV), and taking

$$L_{INT} = \int_{T_{e,det}}^{T_{e,u}} \sqrt{T_e}L_z\, \mathrm{d}T_e \approx m_L(Z, n_e\tau)T_{e,u}, \tag{B2}$$

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

	$n_e\tau = 10^{19}$ m^{-3} ms	$n_e\tau = 10^{20}$ m^{-3} ms	$n_e\tau = 10^{21}$ m^{-3} ms
Nitrogen	2.2×10^{-31}	6.0×10^{-32}	2.2×10^{-32}
Neon	5.6×10^{-31}	1.8×10^{-31}	7.3×10^{-32}
Argon	9.1×10^{-31}	4.9×10^{-31}	3.6×10^{-31}

TABLE 3. Values of m_L in units of Wm^3 eV$^{1/2}$ for different impurities and values of $n_e\tau$ with $n_e = 10^{20}$ m^{-3}, averaging $L_{INT}/T_{e,u}$ over $T_{e,u} = 50$ to 500 eV.

where L_{INT} and the impurity-species-dependent factor $m_L(Z, n_e\tau)$ are considered in Reinke (2017) (representative values of m_L are given in table 3), we can express (B1) directly as

$$c_z = \frac{3.3 \times 10^{-39}}{m_L(\ell_\parallel^*)^{6/7}(1+\kappa^2)^{3/2}(\kappa_z\kappa_0)^{1/7}} \left(\frac{1+\overline{Z}}{\overline{A}}\right)^{1/2} \left(\frac{5}{Z_{eff}+4}\right)^{1/7} \frac{P_{sep}}{B_p f_{GW}^2}. \tag{B3}$$

When expressed in this form, we find that the impurity concentration does not explictly depend on the machine size R_{axis} or a_{geo}. Ignoring for simplicity the turbulence widening, we can write

$$\lambda_{q,u} = 1.6 \frac{a}{R} \frac{\sqrt{m_{DT}\, T_{e,u}}}{e\, B_p} \tag{B4}$$

from Eich *et al.* (2020). Expressing the electron temperature and the power that crosses the separatrix that comes from balancing the loss power given by Spitzer–Härm electron heat conduction in the SOL back into the equation (basically reversing the derivation from Reinke and Goldston (Goldston *et al.* 2017; Reinke 2017)) we arrive at

$$c_z \propto \frac{P_{sep}\sqrt{T}}{n_{sep}^2\, a \lambda_{q,u}} \propto \frac{T^{7/2}\sqrt{T}}{n_{sep}^2\, a^2\, q_{cyl}{}^2} = \frac{T^4\, q_{cyl}^2\, Z_{eff}^2}{n^2\, R^2\, q_{cyl}^4\, Z_{eff}^2}. \tag{B5}$$

Finally recalling the definition of α_t

$$\alpha_t = 3.1 \times 10^{-18} \frac{R\, q_{cyl}^2\, n_e Z_{eff}}{T_e^2} \tag{B6}$$

leads to

$$c_z = \frac{1.4 \times 10^{-24}\sqrt{m_{DT}}(\kappa_0\kappa_z)^{6/7}}{(\ell_\parallel^*)^{13/7} m_L} \left(\frac{1+\overline{Z}}{\overline{A}}\right)^{1/2} \left(\frac{5}{Z_{eff}+4}\right)^{1/7} \frac{q_{cyl}^2\, Z_{eff}^2}{\alpha_t^2}. \tag{B7}$$

Applied to ARC-like parameters: $\ell_\parallel^* \approx 2.4$ (using $L_\parallel = 120$ m), $\kappa_0 = 2300$ Wm$^{-1}eV^{-7/2}$, $\kappa_z = 0.83$ at $Z_{eff} = 1.74$, $\sqrt{\overline{A}/(1+\overline{Z})} \approx 1$ (Goldston *et al.* 2017) and $m_L = 5.7 \times 10^{-31}$ Wm^{3}eV$^{1/2}$ for $n_e\tau = 5 \times 10^{19}$ m^{-3}ms (the value of $n_e\tau$

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

found in Moulton et al. (2021) to give the best match to the Lengyel integral L_{INT} for neon-seeded simulations of ITER)

$$c_z = 2 \times 10^{-5} \cdot \frac{{q_{cyl}}^2 \, Z_{eff}^2}{{\alpha_t}^2}. \tag{B8}$$

For $q_{cyl} = 3.0$ ($q^* = 3.2$), $Z_{eff} = 1.74$, and $\alpha_t = 0.21$, we find c_z=1.2 %. This is slightly larger than the value calculated by the extended Lengyel model of $c_z = 0.9$ %, likely because we have neglected turbulence broadening of $\lambda_{q,u}$.

Appendix C. Nomenclature

Abbreviations and symbols used in this paper, including typical units where applicable.

Species and location identifiers	
$(\cdots)_e$	value for electrons
$(\cdots)_i$	value for ions
$(\cdots)_z$	value for impurity ions
$(\cdots)_{fuel}$	value for fuel ions
$(\cdots)_t$	at the divertor target or sheath-entrance
$(\cdots)_{div}$	in the divertor volume
$(\cdots)_{xpt}$	at the primary X-point
$(\cdots)_u$	at the 'upstream' end of a scrape-off-layer flux tube, typically at the outboard midplane separatrix
$(\cdots)_{sep}$	at the separatrix (often interchangeable with the 'upstream' location)
$\langle\cdots\rangle_{sep}$	averaged over the separatrix
$(\cdots)_{ped}$	in the pedestal (at one half-width towards the core from the center of the pedestal tanh fit)
$(\cdots)_{core}$	in the core, close to the magnetic axis
Acronyms	
UFO	transient impurity event due to dust or debris dislodging from the wall
ELMy	a plasma operational regime with ELMs
MARFE	Multifaceted Asymmetric Radiation From the Edge, strong radiation near the X-point which may be a disruption precursor
SOL	Scrape-Off-Layer
PFC	Plasma-Facing Component
MHD	Magnetohydrodynamics
POPCON	Plasma Operation Contour
SepOS	Separatrix Operational Space
ICRH	Ion Cyclotron Heating
SXD	Super-X Divertor
XPT	X-point target; an advanced divertor configuration with a long, tightly-baffled outer divertor leg and with a secondary X-point near the divertor target.
ELM	Edge-localized modes; ejections of heat and particles when the pedestal reaches a stability limit.
QCE	Quasi-continuous exhaust regime.
RMP	Resonant Magnetic Perturbation
XPR	X-Point Radiator

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

General parameters and terminology

R_{axis}	Major radius of magnetic axis [m]
a_{geo}	Minor radius [m]
I_p	Plasma current [MA]
B_{axis}	Magnetic field on-axis [T]
T_e	electron temperature [eV]
T_i	ion temperature [eV]
Z	charge [$\sim$]
Z_{eff}	effective charge [$\sim$]
B_{pol}	Poloidal magnetic field strength [T]
B_{tor}	Toroidal magnetic field strength [T]
B_{tot}	Total magnetic field strength [T]
$\langle m_i \rangle$	Average ion mass [amu]
$\langle n_e \rangle$	Volume-averaged electron density [$10^{20}/\mathrm{m}^3$]
n_{GW}	Greenwald density limit [$10^{20}/\mathrm{m}^3$]
$f_{Greenwald}$	Ratio of $\langle n_e \rangle / n_{GW}$
P_{fus}	Total fusion power produced [MW]
P_{sep}	Power crossing the separatrix [MW]
P_{L-H}	Threshold value of P_{sep} required to access a high-confinement H-mode [MW]
κ_{areal}	Areal elongation [$\sim$]
κ_{95}	Elongation at the $\Psi_N = 0.95$ surface [$\sim$]
δ_{95}	Triangularity at the $\Psi_N = 0.95$ surface [$\sim$]
q^*	Analytical approximation of q_{95} using equation A10 from Chapter 1 of the 1999 ITER Physics Basis
q_{cyl}	Analytical approximation of q_{95} using equation K.6 from Eich *et al.* (2021) [$\sim$]
q_{95}	Edge safety factor at the $\Psi_N = 0.95$ surface [$\sim$]

Power and particle exhaust parameters

$q_\parallel$	Parallel heat flux density [$\mathrm{GW/m}^2$]
$q_\perp$	Heat flux density perpendicular to the divertor target [$\mathrm{MW/m}^2$]
$\theta_\perp$	Angle of incidence between a magnetic fieldline and the divertor target [°]
α_t	Turbulence parameter giving the relative strength of ballooning instabilities relative to drift waves (2.1) [$\sim$]
$\nu^*_{e,edge}$	Normalized edge collisionality
$L_\parallel$	Parallel connection length from outboard midplane to divertor target, measured $1\lambda_{q,u}$ from the separatrix at the outboard midplane [m]
L_{div}	Parallel length from divertor entrance to divertor target [m]
b_{div}	Divertor broadening factor $\lambda_{INT}/\lambda_{q,u}$ [$\sim$]
λ_{INT}	Integral heat flux width at the divertor target [mm]
S	Gaussian spreading factor [mm]
$\lambda_{q,u}$	Heat flux decay length, measured at the outboard midplane separatrix [mm]
$\rho_{s,pol}$	Sound Larmor radius [mm]
c_z	The concentration of impurity ions relative to electrons []

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

p_{div}	Divertor neutral pressure [Pa]
T_{wall}	Temperature of divertor walls [K]
c_{Ar}	Argon concentration [%]
c_{He}	Helium concentration [%]
$c_{Ar,div}/c_{Ar,core}$	Argon enrichment [∼]
$c_{He,div}/c_{He,core}$	Helium enrichment [∼]
ΔZ_{eff}	Change in the effective charge due to an impurity [∼]
f_{mom}	Fraction of static pressure lost in the scrape-off-layer, [∼]
f_{odiv}	Fraction of P_{sep} directed towards one of the outer divertors [∼]
f_{pow}	Fraction of power lost from a flux-tube in the scrape-off-layer, [∼]
f_{rad}	Fraction of power lost due to radiation from a flux-tube in the scrape-off-layer [∼], $1-\frac{\lambda_{INT}}{\lambda_{q,u}}(1-f_{pow})$
$\Gamma_{i,\perp,t}$	Ion flux perpendicular to the divertor target [$\mathrm{m^{-2}s^{-1}}$]
γ_{sh}	Sheath heat transmission factor [∼]
$\kappa_{e,0}$	Electron heat conductivity prefactor, $\approx 2400\mathrm{W/(eV^{3.5}\cdot m)}/\kappa_z$
κ_z	Correction to $\kappa_{e,0}$ due to finite Z_{eff} from Brown & Goldston (2021)
$\nu^*_{e,ped}$	Normalized pedestal-top collisionality
$\epsilon_{\parallel}$	Parallel energy fluence of a heat-flux transient (2.8) [$\mathrm{MJ/m^2}$]
$\epsilon_{\perp}$	Energy fluence of a heat-flux transient perpendicular to the divertor target (2.12) [$\mathrm{MJ/m^2}$]

REFERENCES

ANGIONI, C. 2025 The impact of tungsten and tungsten transport on H-mode plasmas, experiments and modeling. *Nucl. Fusion* **65**, 062001.

ASAKURA, N. *et al.* 2021 Power exhaust concepts and divertor designs for japanese and european DEMO fusion reactors. *Nucl. Fusion* **61**, 126057.

ASAKURA, N. *et al.* 2023 Recent progress of plasma exhaust concepts and divertor designs for tokamak DEMO reactors. *Nucl. Mater. Energy.* **35**, 101446.

BAUER, W. *et al.* 1979 Alpha transport and blistering in tokamaks. *Nucl. Fusion* **19**, 93–103.

BAUMANN, C. *et al.* 2026 Global tungsten erosion and migration modeling for the EU-DEMO with the ERO2.0 code. *Nucl. Fusion* **66**, 026010.

BERNERT, M. *et al.* 2025 X-point radiation: from discovery to potential application in a future reactor. *Nucl. Mater. Energy* **43**, 101916.

BERNERT, M. *et al.* 2020 X-point radiation, its control and an ELM suppressed radiating regime at the ASDEX upgrade tokamak. *Nucl. Fusion* **61**, 024001.

BERNERT, M. *et al.* 2023 The X-point radiating regime at ASDEX upgrade and TCV. *Nucl. Engng Des./Fusion* **34**, 101376.

BOBKOV, V. *et al.* 2011 ICRF antenna-plasma interactions and its influence on W sputtering in ASDEX upgrade. *J. Nucl. Mater.* **415**, S1005–S1008.

BODY, T., HASSE, C. & CREELY, A. 2023 The SPARC primary reference discharge defined by cfsPOPCON. arXiv [physics.plasm-ph].

BODY, T.A., KALLENBACH, A. & EICH, T. 2025 A simple, accurate model for detachment access. *Nucl. Fusion* **65**, 086002.

BOSCH, H.-S. *et al.* 1997 Particle exhaust studies in ASDEX upgrade. *Plasma Phys. Control. Fusion* **39**, 1771–1792.

BREZINSEK, S. *et al.* 2019 Erosion, screening, and migration of tungsten in the JET divertor. *Nucl. Fusion* **59**, 096035.

BRIDA, D. *et al.* 2025 Transport and profile broadening in the private flux region of ASDEX upgrade and role for power exhaust. *Nucl. Fusion* **65**, 026065.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

BROWN, A.O. & GOLDSTON, R.J. 2021 Generalization of the heuristic drift SOL model for finite collisionality and effect on flow shearing rate vs. interchange growth rate. *Nucl. Engng Des./Fusion* **27**, 101002.

BRUNNER, D. *et al.* 2018*a* The dependence of divertor power sharing on magnetic flux balance in near double-null configurations on alcator C-mod. *Nucl. Fusion* **58**, 076010.

BRUNNER, D., LABOMBARD, B., KUANG, A.Q. & TERRY, J.L. 2018*b* High-resolution heat flux width measurements at reactor-level magnetic fields and observation of a unified width scaling across confinement regimes in the alcator C-mod tokamak. *Nucl. Fusion* **58**, 094002.

BURRELL, K.H. *et al.* 2002 Quiescent H-mode plasmas in the DIII-D tokamak. *Plasma Phys. Control. Fusion* **44**, A253.

BURRELL, K.H. *et al.* 2020 Creation and sustainment of wide pedestal quiescent H-mode with zero net neutral beam torque. *Nucl. Fusion* **60**, 086005.

CAMPBELL, D.J. *et al.* 2025 Introduction and overview: chapter 1 of the special issue: on the path to tokamak burning plasma operation. *Nucl. Fusion* **65**, 093002.

CARPITA, M. *et al.* 2024 Parallel flows as a key component to interpret super-X divertor experiments. *Nucl. Fusion* **64**, 046019.

CHOMICZEWSKA, A. *et al.* 2024 ICRH-related impurity source and control across experiments in H, D, T plasmas at JET-ILW. *Nucl. Fusion* **64**, 076058.

COENEN, J.W. *et al.* 2015 ELM induced tungsten melting and its impact on tokamak operation. *J. Nucl. Mater.* **463**, 78–84.

COWLEY, C. *et al.* 2022 Optimizing detachment control using the magnetic configuration of divertors. *Nucl. Fusion* **62**, 086046.

COWLEY, C., MOULTON, D. & LIPSCHULTZ, B. 2026 Isolating the impacts of divertor closure on fueling and detachment with simplified SOLPS-ITER simulations of MAST-U *Phys. Plasmas* **33**, 022508.

CREELY, A.J. *et al.* 2020 Overview of the SPARC tokamak. *J. Plasma Phys.* **86**, 865860502.

DELABIE, E. *et al.* 2026 Empirical scaling of the L–H threshold power for metal wall tokamaks using a multi-device database. *Nucl. Fusion* **66**, 036016.

DUNNE, M. *et al.* 2024 Quasi-continuous exhaust operational space. *Nucl. Fusion* **64**, 124003.

DUX, R. *et al.* 2009 Plasma-wall interaction and plasma behaviour in the non-boronised all tungsten ASDEX upgrade. *J. Nucl. Mater.* **390–391**, 858–863.

EICH, T. *et al.* 2009 On the asymmetries of ELM divertor power deposition in JET and ASDEX upgrade. *J. Nucl. Mater.* **390–391**, 760–763.

EICH, T. *et al.* 2013 Scaling of the tokamak near the scrape-off layer H-mode power width and implications for ITER. *Nucl. Fusion* **53**, 093031.

EICH, T. *et al.* 2020 Turbulence driven widening of the near-SOL power width in ASDEX upgrade H-mode discharges. *Nucl. Fusion* **60**, 056016.

EICH, T., MANZ, P. & the ASDEX Upgrade Team 2021 The separatrix operational space of ASDEX upgrade due to interchange-drift-alfvén turbulence. *Nucl. Fusion* **61**, 086017.

EICH, T. *et al.* 2017 ELM divertor peak energy fluence scaling to ITER with data from JET, MAST and ASDEX upgrade. *Nucl. Mater. Energy* **12**, 84–90.

EICH, T. *et al.* 2011 Type-I ELM power deposition profile width and temporal shape in JET. *J. Nucl. Mater.* **415**, S856–S859.

ELDON, D., *et al.* 2022 Enhancement of detachment control with simplified real-time modelling on the KSTAR tokamak. *Plasma Phys. Control. Fusion* **64**, 075002.

ELDON, D. *et al.* 2021 An analysis of controlled detachment by seeding various impurity species in high performance scenarios on DIII-D and EAST. *Nucl. Mater. Energy* **27**, 100963.

ERNST, D.R. *et al.* 2024 Broadening of the divertor heat flux profile in high confinement tokamak fusion plasmas with edge pedestals limited by turbulence in DIII-D. *Phys. Rev. Lett.* **132**, 235102.

FAITSCH, M. *et al.* 2025 The quasi-continuous exhaust regime in ASDEX upgrade and JET. *Nucl. Mater. Energy* **42**, 101904.

FAITSCH, M. *et al.* 2023 Analysis and expansion of the quasi-continuous exhaust (QCE) regime in ASDEX upgrade. *Nucl. Fusion* **63**, 076013.

FEDERICI, G. *et al.* 2001 Plasma-material interactions in current tokamaks and their implications for next step fusion reactors. *Nucl. Fusion* **41**, 1967–2137.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

FENSTERMACHER, M.E. *et al.* 2025 Progress in pedestal and edge physics: chapter 3 of the special issue: on the path to tokamak burning plasma operation. *Nucl. Fusion* **65**, 053001.

FIELD, R. *et al.* 2021 Analysis of membrane separation system configurations for the ARC fusion reactor. In *OMC Med Energy Conference and Exhibition*, p. OMC–2021-043.

FIL, A. *et al.* 2020 Separating the roles of magnetic topology and neutral trapping in modifying the detachment threshold for TCV. *Plasma Phys. Control. Fusion* **62**, 035008.

FLANAGAN, J. *et al.* 2025 ELM buffering in the MAST upgrade super-X divertor. *Nucl. Fusion* **65**, 116031.

FUNDAMENSKI, W., PITTS, R.A. & JET EFDA Contributors 2006 A model of ELM filament energy evolution due to parallel losses. *Plasma Phys. Control. Fusion* **48**, 109–156.

GAO, J.M. *et al.* 2023 The effect of impurity seeding into the closed divertor on plasma detachment in the HL-2A tokamak. *Nucl. Fusion* **63**, 036006.

GARCIA, J. *et al.* 2024 Stable deuterium-tritium plasmas with improved confinement in the presence of energetic-ion instabilities. *Nat. Commun.* **15**, 7846.

GOETZ, J.A. *et al.* 2001 The relation between impurity neutral and impurity ion compression in the alcator C-mod divertor. *Nucl. Fusion* **41**, 1751–1754.

GOLDSTON, R.J. 2012 Heuristic drift-based model of the power scrape-off width in low-gas-puff H-mode tokamaks. *Nucl. Fusion* **52**, 013009.

GOLDSTON, R.J. 2015 Theoretical aspects and practical implications of the heuristic drift SOL model. *J. Nucl. Mater.* **463**, 397–400.

GOLDSTON, R.J., REINKE, M.L. & SCHWARTZ, J.A. 2017 A new scaling for divertor detachment. *Plasma Phys. Control. Fusion* **59**, 055015.

GREENWALD, M. *et al.* 2014 20 years of research on the alcator C-mod tokamak. *Phys. Plasmas* **21**, 110501.

GREENWALD, M. *et al.* 1999 Characterization of enhanced high-confinement modes in alcator C-mod. *Phys. Plasmas* **6**, 1943–1949.

GREENWALD, M. *et al.* 1988 A new look at density limits in tokamaks. *Nucl. Fusion* **28**, 2199.

GUILLEMAUT, C. *et al.* 2018 Experimental validation of an analytical kinetic model for edge-localized modes in JET-ITER-like wall. *Nucl. Fusion* **58**, 066006.

HAKOLA, A. *et al.* 2021 Gross and net erosion balance of plasma-facing materials in full-W tokamaks. *Nucl. Fusion* **61**, 116006.

HARRER, G.F. *et al.* 2022 Quasicontinuous exhaust scenario for a fusion reactor: the renaissance of small edge localized modes. *Phys. Rev. Lett.* **129**, 165001.

HENDERSON, S.S. *et al.* 2023 Divertor detachment and reattachment with mixed impurity seeding on ASDEX upgrade. *Nucl. Fusion* **66**, 086024.

HENDERSON, S.S. *et al.* 2024*a* Comparison of reduced model predictions for divertor detachment onset and reattachment timescales in ASDEX upgrade and JET experiments. *Nucl. Fusion* **64**, 066006.

HENDERSON, S.S. *et al.* 2024*b* Validating reduced models for detachment onset and reattachment times on MAST-U. *Nucl. Mater. Energy* **41**, 101765.

HENDERSON, S.S. *et al.* 2021 Parameter dependencies of the experimental nitrogen concentration required for detachment on ASDEX upgrade and JET. *Nucl. Engng Des./Fusion* **28**, 101000.

HENDERSON, S.S. *et al.* 2025 An overview of the STEP divertor design and the simple models driving the plasma exhaust scenario. *Nucl. Fusion* **65**, 016033.

HILLESHEIM, J.C. *et al.* 2026 Overview of the physics basis for the ARC fusion power plant. *J. Plasma Phys.* **92**, E69. DOI: 10.1017/S0022377826101706

HITZLER, F. *et al.* 2020 Impurity transport and divertor retention in Ar and N seeded SOLPS. 5.0 simulations for ASDEX upgrade. *Plasma Phys. Control. Fusion* **62**, 085013.

HOWARD, N.T. *et al.* 2026 Performance and transport in the ARC tokamak. *J. Plasma Phys.* **92**, E67. DOI: 10.1017/S0022377826101573

HU, D. *et al.* 2024 Plasmoid drift and first wall heat deposition during ITER H-mode dual-SPIs in JOREK simulations. *Nucl. Fusion* **64**, 086005.

HU, Q.M. *et al.* 2020 Wide operational windows of edge-localized mode suppression by resonant magnetic perturbations in the DIII-D tokamak. *Phys. Rev. Lett.* **125**, 045001.

HUBBARD, A.E. *et al.* 2001 Pedestal profiles and fluctuations in C-mod enhanced D-alpha H-modes. *Phys. Plasmas* **8**, 2033–2040.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

HUGHES, J.W. *et al.* 2011 Power requirements for superior H-mode confinement on alcator C-mod: experiments in support of ITER. *Nucl. Fusion* **51**, 083007.

IGITKHANOV, Y., DAY, C. & VAROUTIS, S. 2018 A new concept for a higher burn-up fraction improvement in DEMO reactor. *IEEE Trans. Plasma Sci. IEEE Nucl. Plasma Sci. Soc.* **46**, 1466–1470.

KALLENBACH, A. *et al.* 2011 Plasma surface interactions in impurity seeded plasmas. *J. Nucl. Mater.* **415**, S19–S26.

KALLENBACH, A. *et al.* 2015 Partial detachment of high power discharges in ASDEX upgrade. *Nucl. Fusion* **55**, 053026.

KALLENBACH, A. *et al.* 2016 Analytical calculations for impurity seeded partially detached divertor conditions. *Plasma Phys. Control. Fusion* **58**, 045013.

KALLENBACH, A. *et al.* 2024 Divertor enrichment of recycling impurity species (he, N_2, ne, ar, kr) in ASDEX upgrade H-modes. *Nucl. Fusion* **64**, 056003.

KAMADA, Y. *et al.* 2000 Disappearance of giant ELMs and appearance of minute grassy ELMs inJT-60U high-triangularity discharges. *Plasma Phys. Control. Fusion* **42**, A247.

KAPPATOU, A. *et al.* 2025 Overview of the third JET deuterium-tritium campaign. *Plasma Phys. Control. Fusion* **67**, 045039.

KOMM, M. *et al.* 2023 Mitigation of divertor edge localised mode power loading by impurity seeding. *Nucl. Fusion* **63**, 126018.

KOOL, B. *et al.* 2025 Demonstration of super-X divertor exhaust control for transient heat load management in compact fusion reactors. *Nat. Energy* **10** 1116–1131.

KOTSCHENREUTHER, M. *et al.* 2013 Magnetic geometry and physics of advanced divertors: the X-divertor and the snowflake. *Phys. Plasmas* **20**, 102507.

KOTSCHENREUTHER, M. *et al.* 2007 On heat loading, novel divertors, and fusion reactors. *Phys. Plasmas* **14**, 072502.

KRASHENINNIKOV, S.I. & KUKUSHKIN, A.S. 2017 Physics of ultimate detachment of a tokamak divertor plasma. *J. Plasma Phys.* **83**, 155830501.

KRIEGER, K. *et al.* 2025 Scrape-off layer and divertor physics: chapter 5 of the special issue: on the path to tokamak burning plasma operation. *Nucl. Fusion* **65**, 043001.

KUANG, A.Q. *et al.* 2020 Divertor heat flux challenge and mitigation in SPARC. *J. Plasma Phys.* **86**, 865860505

KUANG, A.Q. *et al.* 2018 Conceptual design study for heat exhaust management in the ARC fusion pilot plant. *Fusion Engng Des.* **137**, 221–242.

LABIT, B. *et al.* 2019 Dependence on plasma shape and plasma fueling for small edge-localized mode regimes in TCV and ASDEX upgrade*. *Nucl. Fusion* **59**, 086020.

LABOMBARD, B. *et al.* 2005 Evidence for electromagnetic fluid drift turbulence controlling the edge plasma state in the alcator C-mod tokamak. *Nucl. Fusion* **45**, 1658.

LABOMBARD, B. *et al.* 2013 X-point target divertor concept and the alcator DX high power divertor test facility. *Bull. Am. Phys. Soc.* **2013**, CO4.002.

LABOMBARD, B. *et al.* 2015 ADX: a high field, high power density, advanced divertor and RF tokamak. *Nucl. Fusion* **55**, 053020.

LANG, P.T. *et al.* 2004 ELM pace making and mitigation by pellet injection in ASDEX upgrade. *Nucl. Fusion* **44**, 665–677.

LASA, A. *et al.* 2026 Sub-Surface Gas Dynamics in the Arc Divertor under Prolonged Exposure to Detached Plasma Conditions. *Fusion Sci. Technol.*, 1–15

LEE, K. *et al.* 2025 X -point target radiator regime in tokamak divertor plasmas. *Phys. Rev. Lett.* **134**, 185102.

LENGYEL, L.L. 1981 Analysis of radiating plasma boundary layers. Tech. Rep., Max Planck Institute for Plasma Physics.

LI, K. *et al.* 2021 Comparison of divertor behavior and plasma confinement between argon and neon seeding in EAST. *Nucl. Fusion* **61**, 066013.

LIANG, Y. *et al.* 2007 Active control of type-I edge localized modes on JET. *Plasma Phys. Control. Fusion* **49**, B581–B589.

LINKE, J. *et al.* 2019 Challenges for plasma-facing components in nuclear fusion. *Matter Radiat. Extrem.* **4**, 056201.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

LIPSCHULTZ, B. 1987 Review of MARFE phenomena in tokamaks. *J. Nucl. Mater.* **145–147**, 15–25.
LIPSCHULTZ, B., PARRA, F.I. & HUTCHINSON, I.H. 2016 Sensitivity of detachment extent to magnetic configuration and external parameters. *Nucl. Fusion* **56**, 056007.
LOARTE, A. *et al.* 2003 Characteristics of type I ELM energy and particle losses in existing devices and their extrapolation to ITER. *Plasma Phys. Control. Fusion* **45**, 1549–1569.
LOGAN, N.C. *et al.* 2025 Metrics and extrapolation of resonant magnetic perturbation thresholds for ELM suppression. *Nucl. Fusion* **65**, 076029.
MAGET, P. *et al.* 2025 Tungsten control in long pulse operation: feedback from WEST to ITER. *Plasma Phys. Control. Fusion* **67**, 045005.
MANZ, P. 2025 Density and radiation operational limits to H-mode sustainment. *Nucl. Fusion* **65**, 092001.
MANZ, P., EICH, T. & GROVER, O. 2023 The power dependence of the maximum achievable H-mode and (disruptive) L-mode separatrix density in ASDEX upgrade. *Nucl. Fusion* **63**, 076026.
MARTIN, Y.R., T., TAKIZUKA & the ITPA CDBM H-Mode Threshold Database Working Group 2008 Power requirement for accessing the H-mode in ITER. *J. Phys. Conf. Ser.* **123**, 012033.
MASLINE, R., WIGRAM, M. & WHYTE, D. 2025 Modeling helium compression and enrichment in DIII-D edge plasmas using the SOLPS-ITER code. arXiv [physics.plasm-ph].
MAURIZIO, R. *et al.* 2024 Experiments on plasma detachment in a V-shaped slot divertor in the DIII-D tokamak. *Nucl. Fusion* **64**, 086048.
MENG, L.Y. *et al.* 2024 Effects of divertor geometry on detachment and core plasma performance with impurity seeding in EAST. *Nucl. Fusion* **64**, 126045.
MILLER, M.A. *et al.* 2025 Determination of confinement regime boundaries via separatrix parameters on alcator C-mod based on a model for interchange-Drift-Alfvén turbulence. *Nucl. Fusion* **65**, 052002.
MOULTON, D., *et al.* 2021 Comparison between SOLPS-4.3 and the lengyel model for ITER baseline neon-seeded plasmas. *Nucl. Fusion* **61**, 046029.
MOYER, R.A. *et al.* 2005 Edge localized mode control with an edge resonant magnetic perturbation. *Phys. Plasmas* **12**, 056119.
NEU, R. *et al.* 2026 Review on ASDEX Upgrade operation with tungsten plasma facing components. *Nucl. Mater. Energy* **46**, 102063.
NILLE, D. *et al.* 2019 Analytic 1D approximation of the divertor broadening S in the divertor region for conductive heat transport. *Plasma Phys. Control. Fusion* **61**, 085016.
OSAWA, R.T. *et al.* 2024 Assessment of the impact of fuelling puff location on divertor impurity compression and enrichment in STEP. *Nucl. Fusion* **64**, 106007.
OSTUNI, V. *et al.* 2022 Core radiative collapse characterisation and integrated modelling in WEST plasmas. *Nucl. Fusion* **62**, 106034.
PAZ-SOLDAN, C. *et al.* 2024 Plasma performance and operational space with an RMP-ELM suppressed edge. *Nucl. Fusion* **64**, 096004.
PERILLO, R. *et al.* 2025 First-wall fluxes from large and small ELMs. *Phys. Plasmas* **32**, 022501.
PERILLO, R. *et al.* 2026 Pedestal origin and extrapolation of high-density small edge-localised-modes peak parallel energy fluence in ITER and SPARC. *J. Plasma Phys.* **92**, E35
PIRAS, F. *et al.* 2009 Snowflake divertor plasmas on TCV. *Plasma Phys. Control. Fusion* **51**, 055009.
PITCHER, C.S. & STANGEBY, P.C. 1997 Experimental divertor physics. *Plasma Phys. Control. Fusion* **39**, 779–930.
PÜTTERICH, T., *et al.* 2019 Determination of the tolerable impurity concentrations in a fusion reactor using a consistent set of cooling factors. *Nucl. Fusion* **59**, 056013.
RAJ, H. *et al.* 2022 Improved heat and particle flux mitigation in high core confinement, baffled, alternative divertor configurations in the TCV tokamak. *Nucl. Fusion* **62**, 126035.
RAVENSBERGEN, T. *et al.* 2021 Real-time feedback control of the impurity emission front in tokamak divertor plasmas. *Nat. Commun.* **12**, 1105.
REDL, A. *et al.* 2023 Energy load on first wall components in high density, small ELM regimes in ASDEX upgrade. *Nucl. Mater. Energy* **34**, 101319.
REIMERDES, H. *et al.* 2021 Initial TCV operation with a baffled divertor. *Nucl. Fusion* **61**, 024002.
REINKE, M.L. 2017 Heat flux mitigation by impurity seeding in high-field tokamaks. *Nucl. Fusion* **57**, 034004.
REITER, D. *et al.* 1991 Helium removal from tokamaks. *Plasma Phys. Control. Fusion* **33**, 1579–1600.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

RIMINI, F.G., JET Contributors & the EUROfusion Tokamak Exploitation Team 2025 40 years of JET operations: a unique contribution to fusion science. *Plasma Phys. Control. Fusion* **67**, 033001.

RIVALS, N. *et al.* 2022 SOLEDGE3X full vessel plasma simulations for computation of ITER first-wall fluxes. *Contrib. Plasma Phys.* **62**, e202100182.

RYUTOV, D.D. 2007 Geometrical properties of a 'snowflake' divertor. *Phys. Plasmas* **14**, 064502.

SAIBENE, G. *et al.* 2005 Characterization of small ELM experiments in highly shaped single null and quasi-double-null plasmas in JET. *Nucl. Fusion* **45**, 297–317.

SAKASAI, A. *et al.* 1999 Helium exhaust in ELMy H-mode plasmas with W-shaped pumped divertor of JT-60U. *J. Nucl. Mater.* **266–269**, 312–317.

SCARABOSIO, A. *et al.* 2015 Scaling of the divertor power spreading (S-factor) in open and closed divertor operation in JET and ASDEX upgrade. *J. Nucl. Mater.* **463**, 49–54.

SCOTTO D'ABUSCO, M. *et al.* 2025 3D modelling of n = 1 RMP driven heat fluxes on the SPARC tokamak PFCs using HEAT. *Nucl. Fusion* **65**, 096030.

SENICHENKOV, I.Y. *et al.* 2019 On mechanisms of impurity leakage and retention in the tokamak divertor. *Plasma Phys. Controlled Fusion* **61**, 045013.

SIEGLIN, B. *et al.* 2016 Investigation of scrape-off layer and divertor heat transport in ASDEX upgrade L-mode. *Plasma Phys. Control. Fusion* **58**, 055015.

SNYDER, P.B. *et al.* 2011 A first-principles predictive model of the pedestal height and width: development, testing and ITER optimization with the EPED model. *Nucl. Fusion* **51**, 103016.

SNYDER, P.B. *et al.* 2012 The EPED pedestal model and edge localized mode-suppressed regimes: studies of quiescent H-mode and development of a model for edge localized mode suppression via resonant magnetic perturbations. *Phys. Plasmas* **19**, 056115.

SNYDER, P.B. *et al.* 2002 Edge localized modes and the pedestal: a model based on coupled peeling-ballooning modes. *Phys. of Plasmas* **9**, 2037.

SOLANO, E.R. *et al.* 2021 L–H transition threshold studies in helium plasmas at JET. *Nucl. Fusion* **61**, 124001.

SONG, Y. *et al.* 2023 Realization of thousand-second improved confinement plasma with super I-mode in tokamak EAST. *Sci. Adv.* **9**, eabq5273.

SOUKHANOVSKII, V.A. 2017 A review of radiative detachment studies in tokamak advanced magnetic divertor configurations. *Plasma Phys. Control. Fusion* **59**, 064005.

SOUKHANOVSKII, V.A. *et al.* 2011 Taming the plasma–material interface with the 'snowflake' divertor in NSTX. *Nucl. Fusion* **51**, 012001.

STANGEBY, P.C. 2000 *The Plasma Boundary of Magnetic Fusion Devices*. Institute of Physics Publishing.

STANGEBY, P.C. 2018 Basic physical processes and reduced models for plasma detachment. *Plasma Phys. Control. Fusion* **60**, 044022.

STANGEBY, P.C. *et al.* 2022 Developing solid-surface plasma facing components for pilot plants and reactors with replenishable wall claddings and continuous surface conditioning. part a: concepts and questions. *Plasma Phys. Control. Fusion* **64**, 055018.

STOBER, J. *et al.* 2001 Type II ELMy H modes on ASDEX upgrade with good confinement at high density. *Nucl. Fusion* **41**, 1123–1134.

SUN, H.J. *et al.* 2025 Impact of the plasma boundary on machine operation and the risk mitigation strategy on JET. *Nucl. Fusion* **65**, 076012.

THEILER, C. *et al.* 2017 Results from recent detachment experiments in alternative divertor configurations on TCV. *Nucl. Fusion* **57**, 072008.

TOBITA, K. *et al.* 2003 First wall issues related with energetic particle deposition in a tokamak fusion power reactor. *Fusion Engng Des.* **65**, 561–568.

UEDA, Y. *et al.* 2014 Research status and issues of tungsten plasma facing materials for ITER and beyond. *Fusion Engng Des.* **89**, 901–906.

UMANSKY, M.V. *et al.* 2017 Attainment of a stable, fully detached plasma state in innovative divertor configurations. *Phys. Plasmas* **24**, 056112.

UMANSKY, M.V. *et al.* 2020 Study of passively stable, fully detached divertor plasma regimes attained in innovative long-legged divertor configurations. *Nucl. Fusion* **60**, 016004.

VALANJU, P.M., KOTSCHENREUTHER, M., MAHAJAN, S.M. & CANIK, J. 2009 Super-X divertors and high power density fusion devices. *Phys. Plasmas* **16**, 056110.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press

VAN ROOIJ, G.J. *et al.* 2013 Tungsten divertor erosion in all metal devices: lessons from the ITER like wall of JET. *J. Nucl. Mater.* **438**, S42–S47.

VERBEEK, H. *et al.* 1998 Interaction of charge exchange neutrals with the main chamber walls of plasma machines. *Nucl. Fusion* **38**, 1789–1803.

VERHAEGH, K. *et al.* 2025 Divertor shaping with neutral baffling as a solution to the tokamak power exhaust challenge. *Commun. Phys.* **8**, 1–15.

VICTOR, B.S. *et al.* 2020 Impurity transport in the pedestal of H-mode plasmas with resonant magnetic perturbations. *Plasma Phys. Control. Fusion* **62**, 095021.

VIEZZER, E. 2018 Access and sustainment of naturally ELM-free and small-ELM regimes. *Nucl. Fusion* **58**, 115002.

VINCENZI, P. *et al.* 2025 Connecting recent JET isotope L–H transition studies to H-mode access in new ITER scenarios. *Plasma Phys. Control. Fusion* **67**, 045013.

WADE, M.R. *et al.* 1995 Helium transport and exhaust studies in enhanced confinement regimes in DIII-D. *Phys. Plasmas* **2**, 2357–2365.

WADE, M.R. *et al.* 1998 Impurity enrichment studies with induced scrape-off layer flow on DIII-D. *Nucl. Fusion* **38**, 1839–1859.

WADE, M.R. *et al.* 2015 Advances in the physics understanding of ELM suppression using resonant magnetic perturbations in DIII-D. *Nucl. Fusion* **55**, 023002.

WAGNER, F. 2007 A quarter-century of H-mode studies. *Plasma Phys. Control. Fusion* **49**, B1.

WAGNER, F. *et al.* 1982 Regime of improved confinement and high beta in neutral-beam-heated divertor discharges of the ASDEX tokamak. *Phys. Rev. Lett.* **49**, 1408–1412.

WANG, L. *et al.* 2021 Integration of full divertor detachment with improved core confinement for tokamak fusion plasmas. *Nat. Commun.* **12**, 1365.

WANG, L. *et al.* 2022 Achievements of actively controlled divertor detachment compatible with sustained high confinement core in DIII-D and EAST. *Nucl. Fusion* **62**, 076002.

WEST, W.P. *et al.* 1999 Impurity enrichment and radiative enhancement using induced SOL flow in DIII-D. *J. Nucl. Mater.* **266–269**, 44–50.

WHYTE, D.G. *et al.* 2010 I-mode: an H-mode energy confinement regime with L-mode particle transport in alcator C-mod. *Nucl. Fusion* **50**, 105005.

WIGRAM, M.R.K. *et al.* 2019 Performance assessment of long-legged tightly-baffled divertor geometries in the ARC reactor concept. *Nucl. Fusion* **59**, 106052.

WILSON, H.R. *et al.* 2006 Magneto-hydrodynamic stability of the H-mode transport barrier as a model for edge localized modes: an overview. *Plasma Phys. Control. Fusion* **48**, A71–A84.

WOLFRUM, E. *et al.* 2011 Characterization of edge profiles and fluctuations in discharges with type-II and nitrogen-mitigated edge localized modes in ASDEX upgrade. *Plasma Phys. Control. Fusion* **53**, 085026.

XU, G.S. *et al.* 2020 Divertor impurity seeding with a new feedback control scheme for maintaining good core confinement in grassy-ELM H-mode regime with tungsten monoblock divertor in EAST. *Nucl. Fusion* **60**, 086001.

YOSHIDA, N. & HIROOKA, Y. 1998 Impacts of charge-exchange neutrals on degradation of plasma-facing materials. *J. Nucl. Mater.* **258–263**, 173–182.

ZHENG, G.Y., XU, X.Q., RYUTOV, D.D., PAN, Y.D. & XIA, T.Y. 2014 Magnetic configuration flexibility of snowflake divertor for HL-2M. *Fusion Engng Des.* **89**, 2621–2627.

ZITO, A. *et al.* 2023 Investigation of helium exhaust dynamics at the ASDEX upgrade tokamak with full-tungsten wall. *Nucl. Fusion* **63**, 096027.

ZITO, A. *et al.* 2025 SOLPS-ITER modelling of helium transport, recycling and pumping at the ASDEX upgrade tokamak. *Nucl. Fusion* **65**, 046022.

ZOHM, H. *et al.* 2021 The EU strategy for solving the DEMO exhaust problem. *Fusion Engng Des.* **166**, 112307.

ZOHM, H. *et al.* 2024 Overview of ASDEX upgrade results in view of ITER and DEMO. *Nucl. Fusion* **64**, 112001.

ZOHM, H. 1996 Edge localized modes (ELMs). *Plasma Phys. Control. Fusion* **38**, 105–128.

https://doi.org/10.1017/S002237782610155X Published online by Cambridge University Press